Lab 4:

MOS-Capacitors and SOI Devices – Quantum Effects

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Part 1: MOS Capacitors Modeling – CV curves and Quantum effects

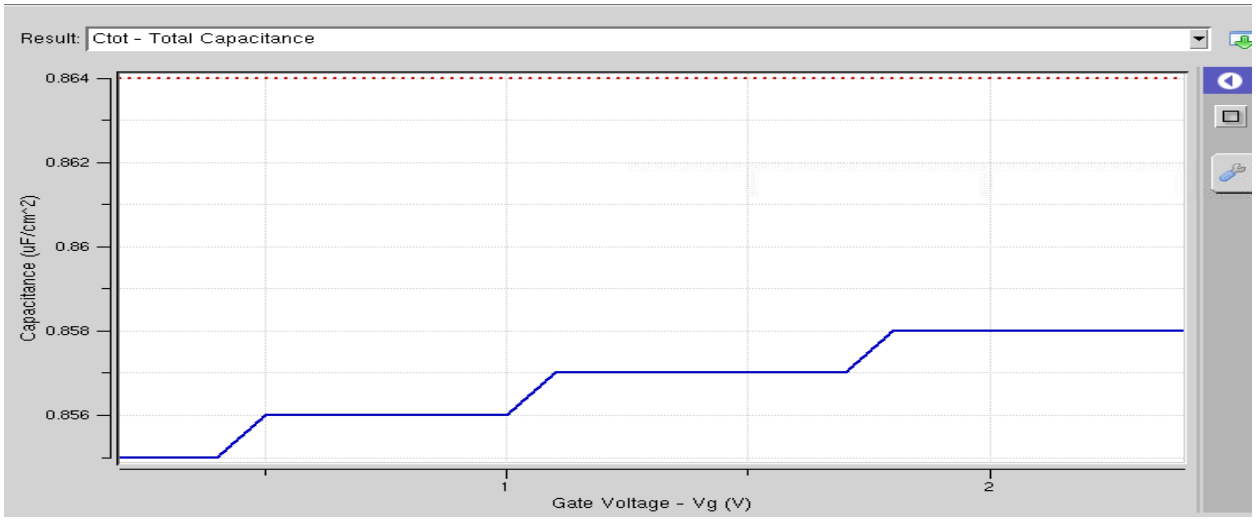
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1. *(2.4 points)* Run Simulation setups 1, 2 and 3 and answer the following questions:

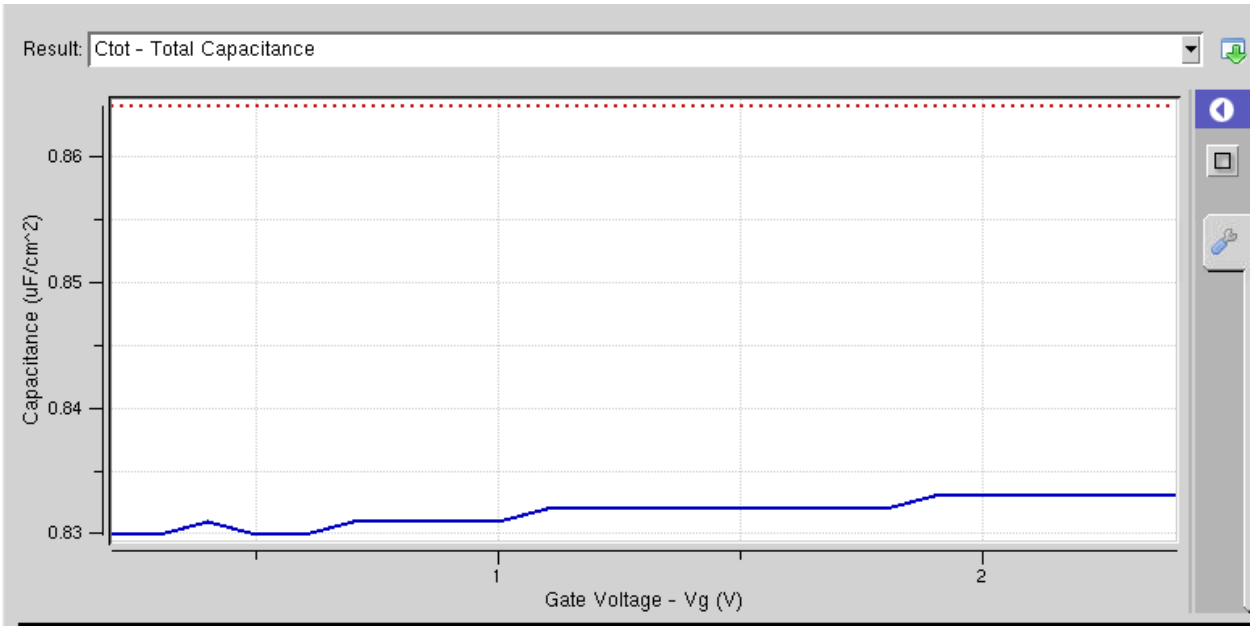
A. Plot the total gate capacitance as a function of the gate voltage. For the gate voltage of 2V what is the degradation of the total gate capacitance when using semiclassical model with Maxwell-Boltzmann and Fermi-Dirac statistics, and quantum-mechanical charge description model?

Answer: I plotted the total gate capacitance (Ctotal) versus the gate voltage for all three simulation setups. To find the degradation at a gate voltage of 2V, I compared the simulated Ctotal value to the ideal oxide capacitance (Cox), which my results showed to be 0.864 µF/cm² for all runs. In Simulation 1 (Semiclassical, Maxwell-Boltzmann), the Ctotal at 2V was 0.858 µF/cm², resulting in a very small degradation of only 0.69%. In Simulation 2 (Semiclassical, Fermi-Dirac), the Ctotal dropped to 0.833 µF/cm², which is a more noticeable degradation of 3.59%. Finally, in Simulation 3 (Quantum-Mechanical), the Ctotal was the lowest at 0.803 µF/cm², showing the largest degradation by far at 7.06%. This clearly demonstrates that the quantum-mechanical model predicts the most significant capacitance degradation, is due to the "charge set-back" effect. This effect pushes the inversion charge peak away from the interface, creating an additional inversion layer capacitance in series with the oxide capacitance and thus lowering the total Ctotal.

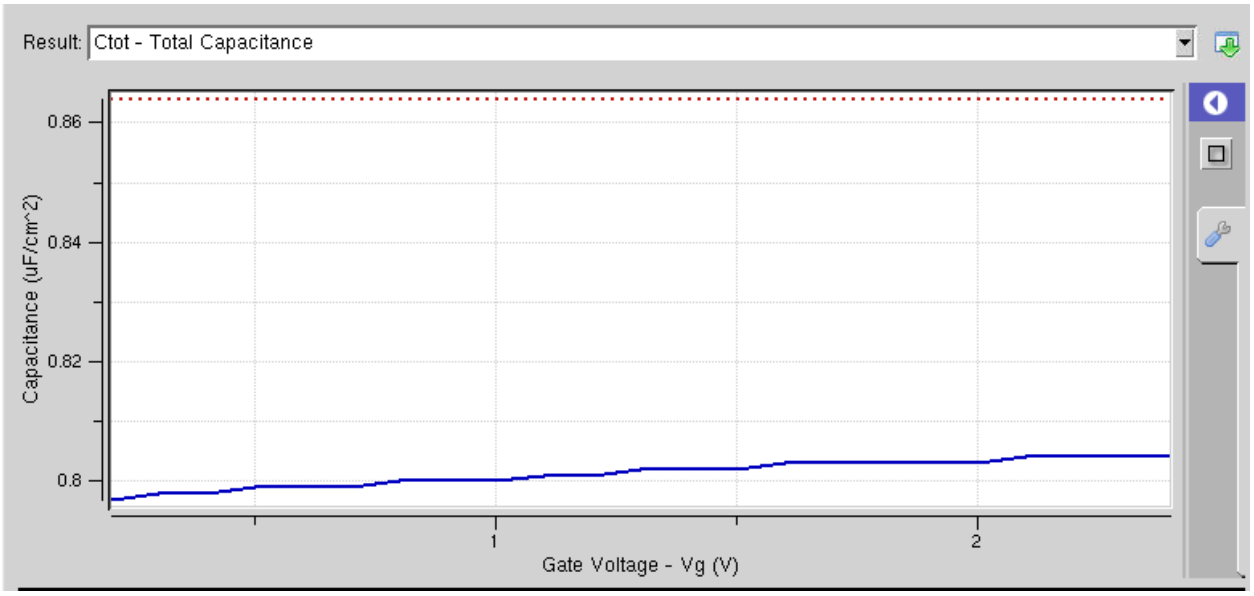
Simulation 1:



Simulation 2:



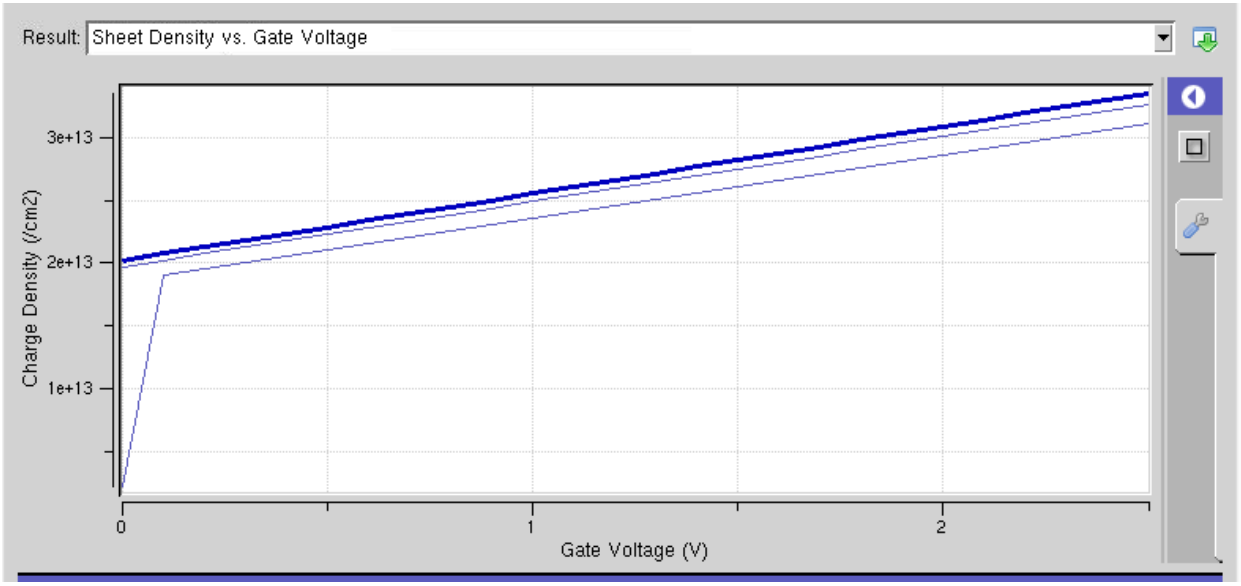
Simulation 3:



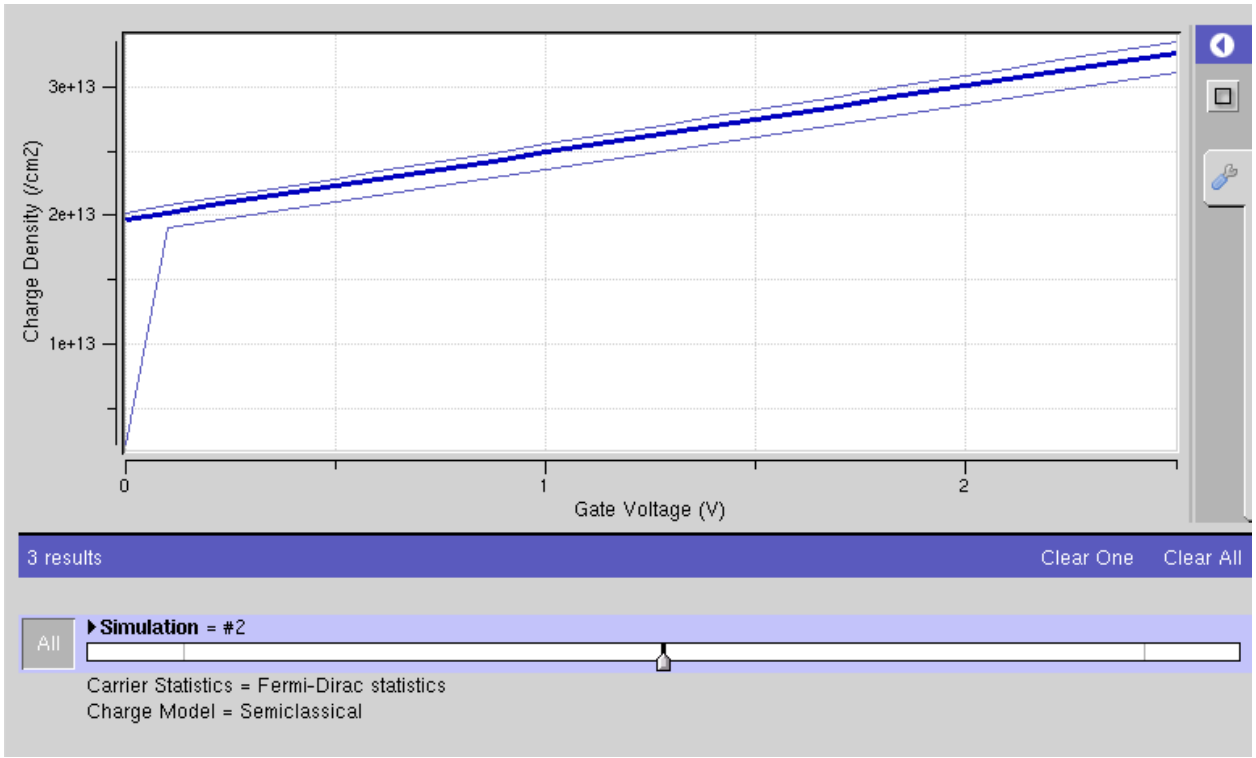
B. Plot the electron sheet density as a function of gate voltage for the three simulation setups. What can you say about the sheet density variation (and the threshold voltage) for the three different charge description models?
Answer: I plotted the electron sheet density as a function of gate voltage for the three simulation setups. The plot shows that for all three models, the sheet density increases linearly with gate voltage over the 0V to 2.5V range. This indicates that the threshold voltage for all three models is below 0V, and my simulations were conducted entirely within the strong inversion regime. For any given gate voltage in this range, the Semiclassical

(Maxwell-Boltzmann) model predicts the highest sheet density (3.36 x 10^{13} /cm² at 2.5V), followed by the Semiclassical (Fermi-Dirac) model (3.27 x 10^{13} /cm² at 2.5V). The Quantum-Mechanical model consistently predicts the lowest sheet density (3.11 x 10^{13} /cm² at 2.5V). This data also shows that the quantum-mechanical model will have the highest (or least negative) threshold voltage, as it requires more gate drive to achieve the same level of inversion charge. Quantum confinement raises the energy of the ground-state subband, thus increasing the threshold voltage.

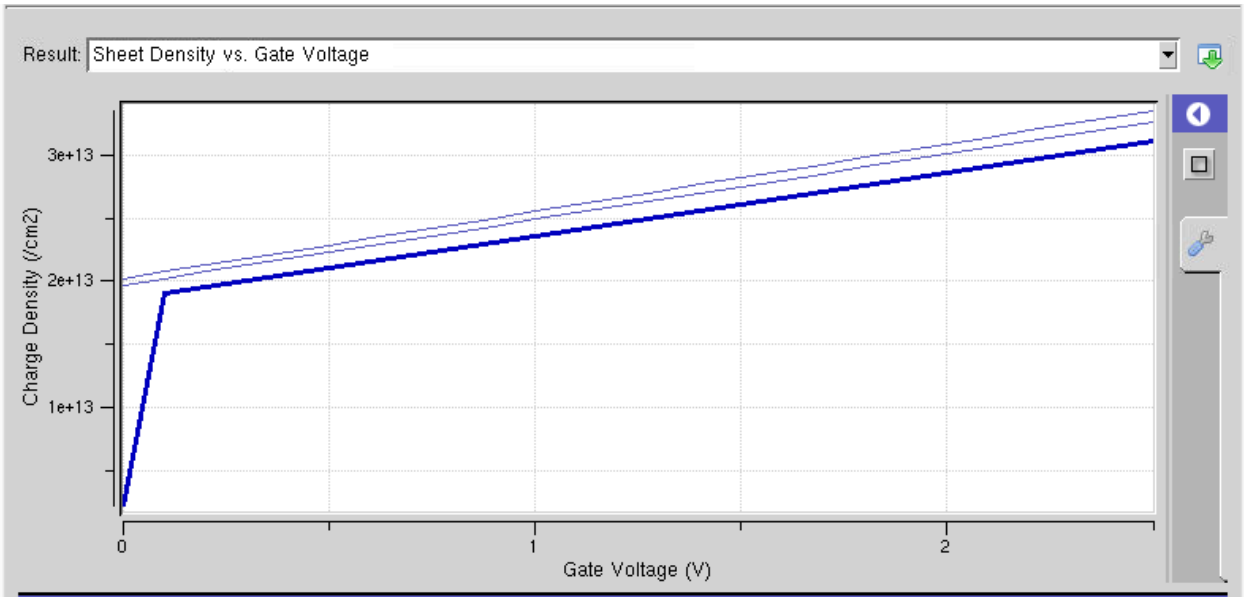
Simulation 1:



Simulation 2:



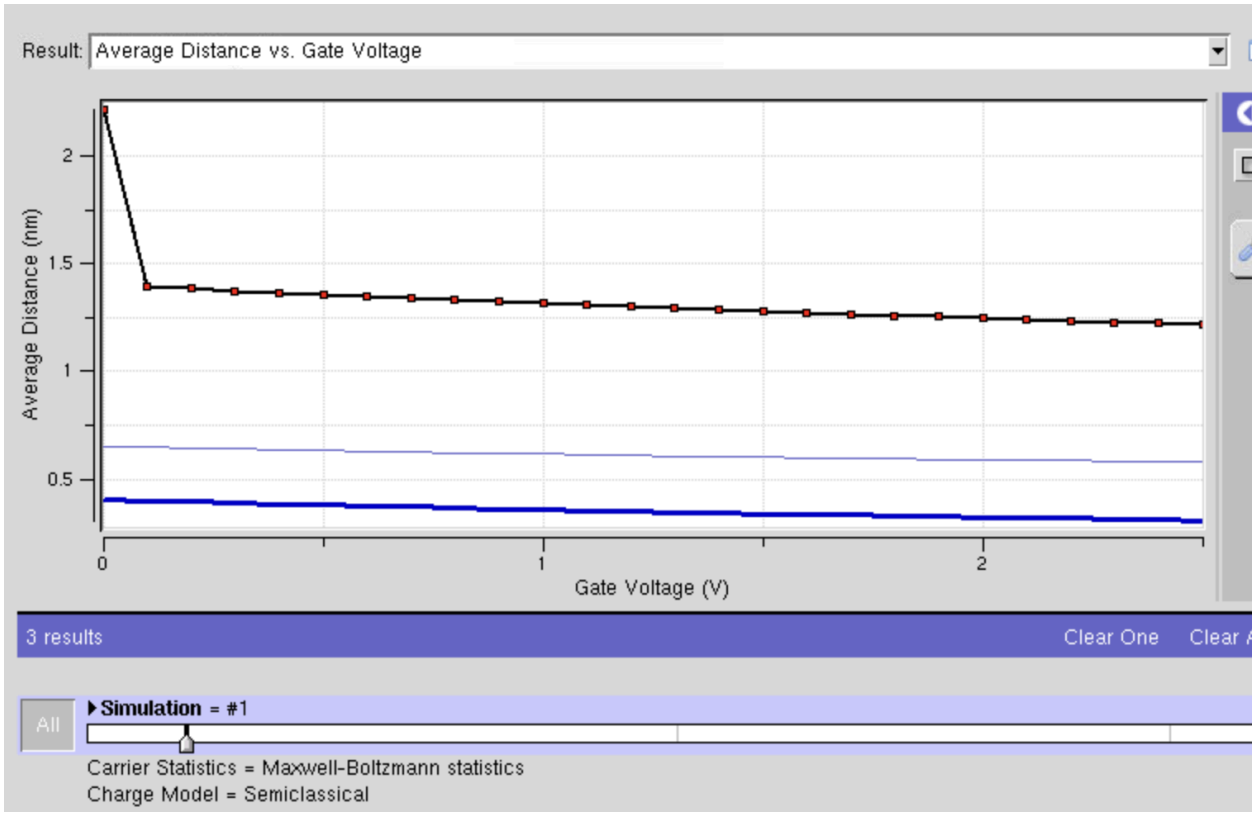
Simulation 3:



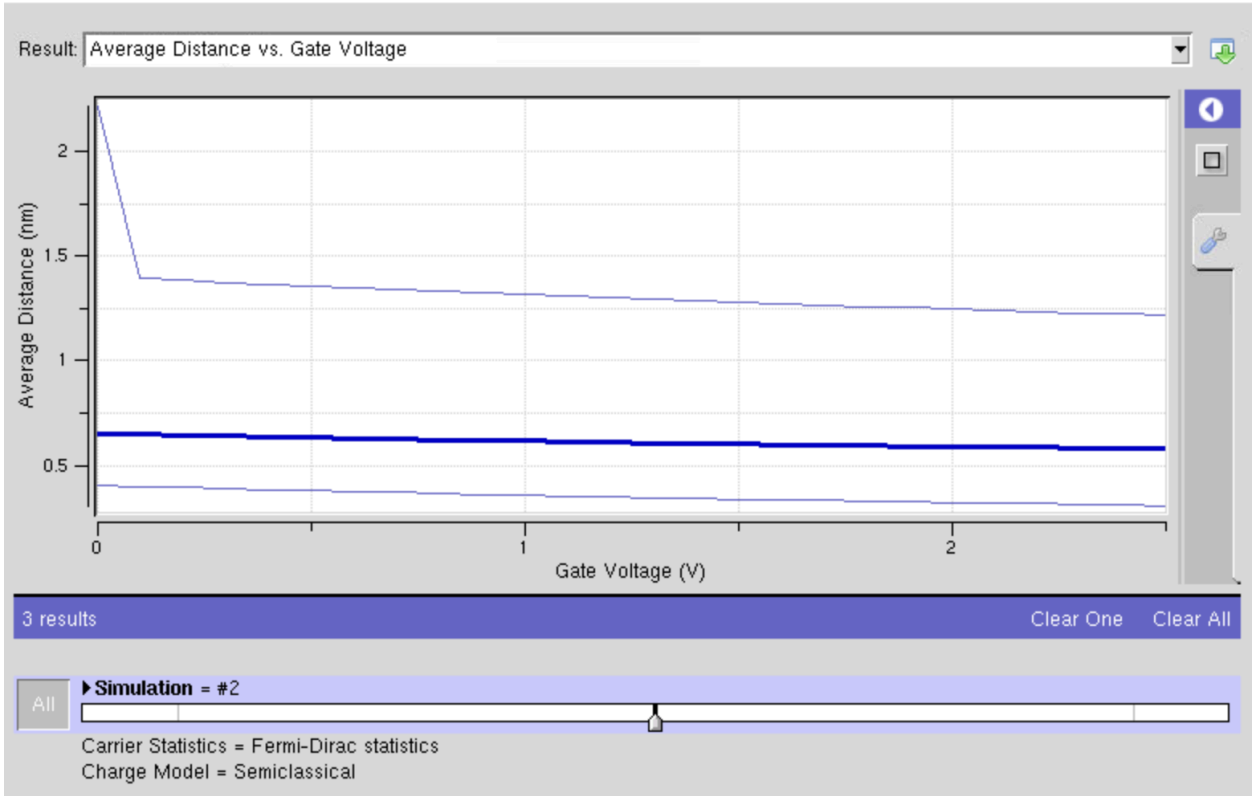
C. Plot the average distance of the electrons from the interface for the three charge description models. Given the gate voltage of 2V, under what conditions we have minimum average distance of the electrons from the interface. Why?

Answer: Based on my simulation data, at a gate voltage of 2V, the Semiclassical (Maxwell-Boltzmann) model (Simulation 1) gives the minimum average distance, which is 0.324 nm. This is much smaller than the Semiclassical (Fermi-Dirac) model's 0.591 nm and the Quantum-Mechanical model's 1.247 nm. The reason this model gives the smallest distance is that semiclassical models assume the electron charge density peaks *at* the semiconductor-oxide interface, rather than away from it . The quantum-mechanical model, which is more physically accurate, accounts for the electron wavefunction vanishing at the oxide barrier, which forces the charge peak to be "set back" from the interface, resulting in a significantly larger average distance . The semiclassical model ignores this quantum effect, leading to a much smaller calculated average distance.

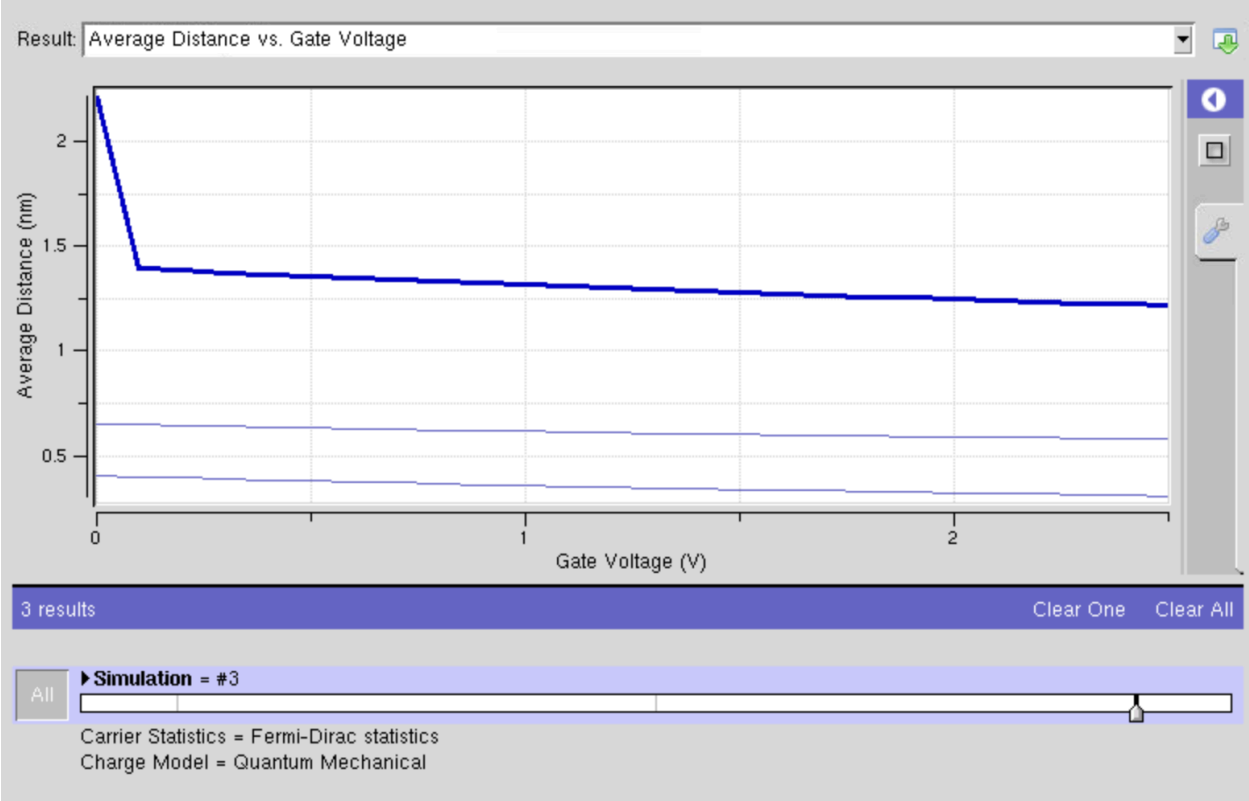
Simulation 1:



Simulation 2:



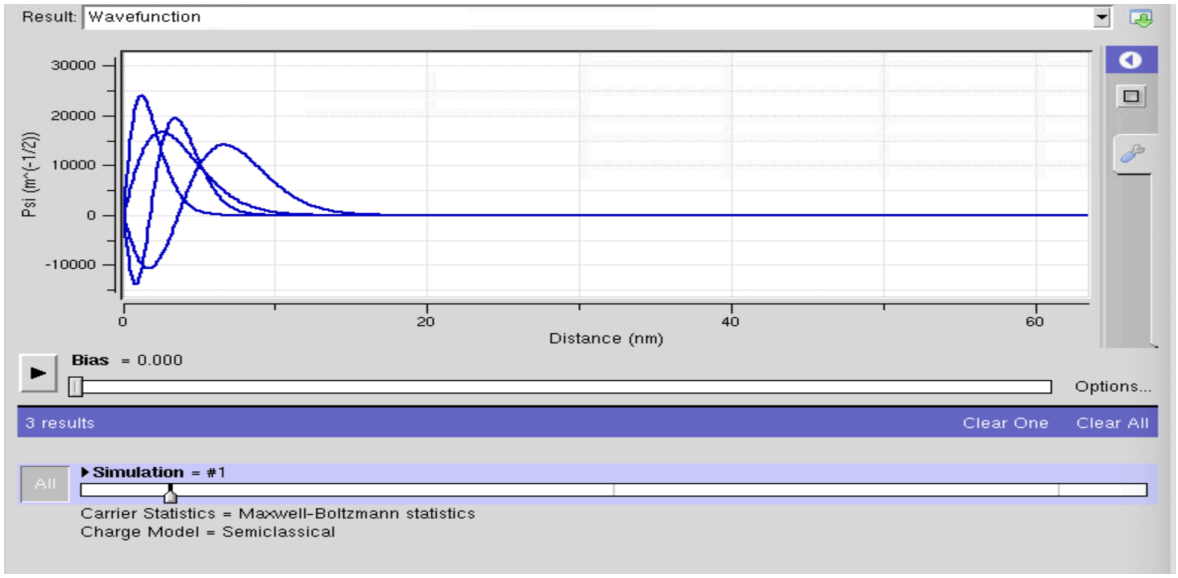
Simulation 3:



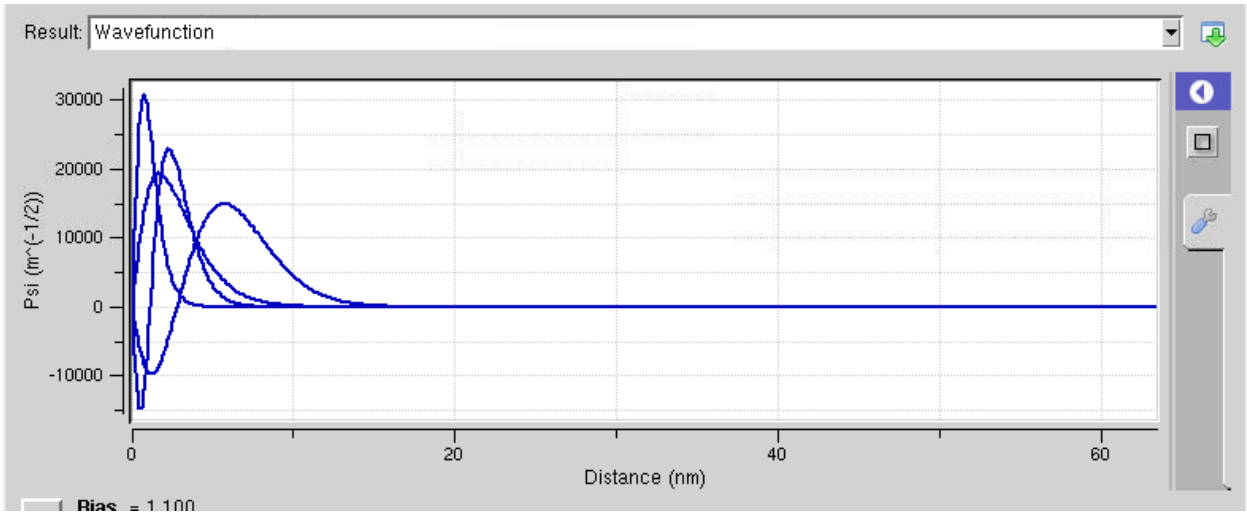
D. For the quantum-mechanical charge description model, plot the unprimed and primed wavefunctions for V_G = 2V. What can you say about the spatial extent of these wavefunctions for the electrons sitting in the unprimed or primed ladder of subbands?

Answer: I have analyzed the wavefunction data from my Simulation 3 (Quantum-Mechanical model). This data is divided into two sets, the unprimed (psi1x) and primed (psi2x) ladders, which correspond to the Delta-2 and Delta-4 bands, respectively. Based on these results, I can state that the wavefunctions for the electrons in the primed (Delta-4) ladder of subbands have a significantly larger spatial extent and are less confined than those in the unprimed (Delta-2) ladder. My data clearly show this: the ground state of the unprimed ladder (psi11) peaks at approximately 1.5 to 1.65 nm and is confined very close to the interface, decaying to almost zero by 15 nm. In contrast, the ground state of the primed ladder (psi21) peaks much further into the substrate, at about 5.4 nm, and is far more spread out, extending well beyond 20 nm. This difference is due to the different effective masses for each ladder perpendicular to the interface. The unprimed ladder has a heavy perpendicular mass, which leads to stronger confinement and lower energy levels. The primed ladder has a light perpendicular mass, resulting in weaker confinement, higher energy levels, and wavefunctions that penetrate much further into the silicon.

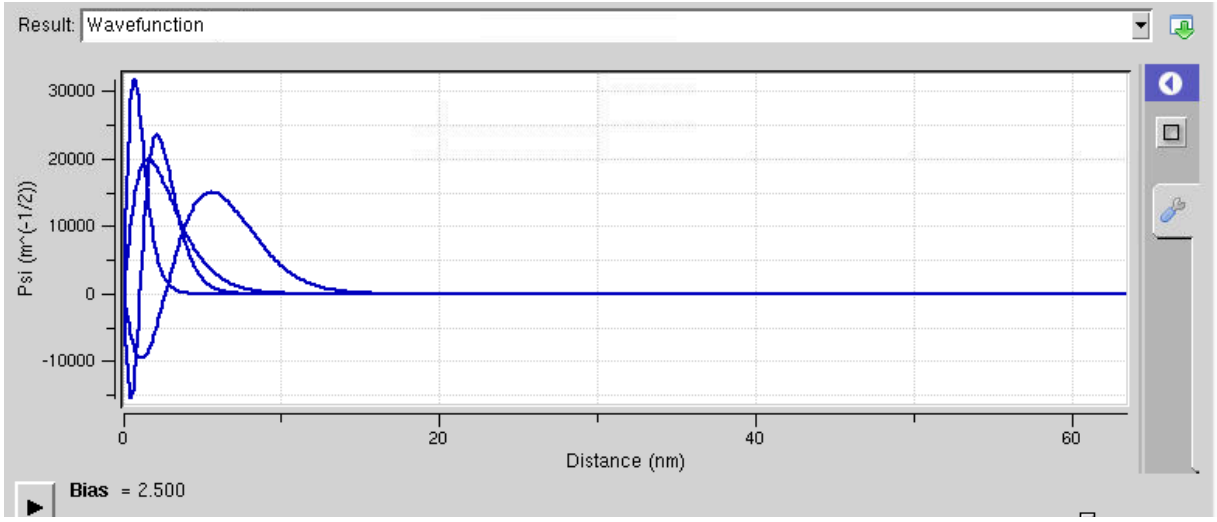
Simulation 1:



Simulation 2:



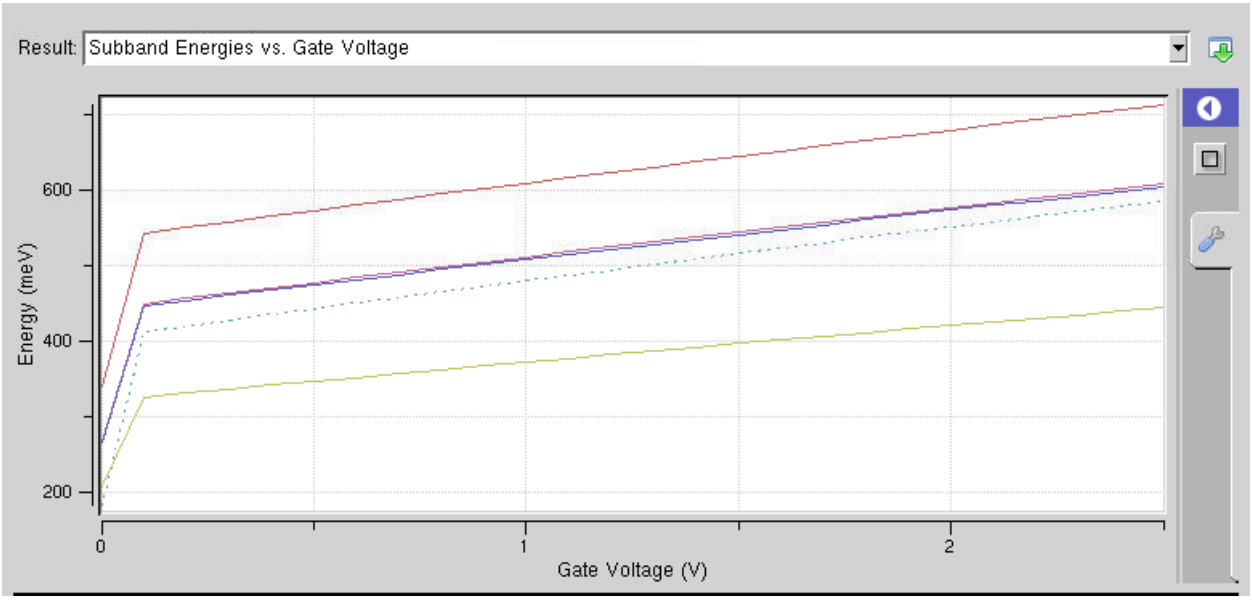
Simulation 3:



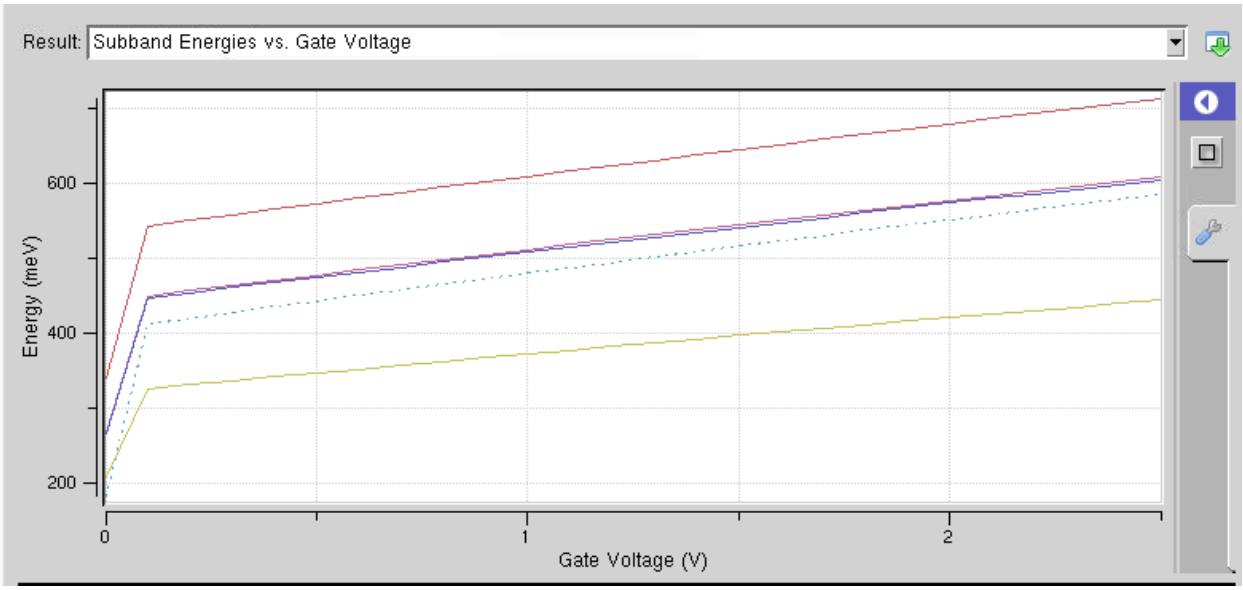
E. For the quantum-mechanical charge description model, plot the subband energies vs gate voltage? What happens with the subband energies and the subband separation when increasing the gate voltage? Why?

I have analyzed the subband energy data from my quantum mechanical simulation (Simulation 3). Based on this data, as the gate voltage increases from 0V to 2.5V, all of the subband energies increase. For example, the lowest subband, E11, increases from 267.17 meV to 605.41 meV, and the next subband, E21, increases from 340.65 meV to 714.34 meV. The subband separation also increases as the gate voltage increases. I can see this by comparing the energy gap between two subbands at different voltages. For instance, the separation between E21 and E11 is about 96.3 meV at a gate voltage of 0.1V (543.65 meV - 447.38 meV), but at 2.5V, this separation grows to 108.9 meV (714.34 meV - 605.41 meV). This happens because increasing the gate voltage creates a stronger electric field, which makes the triangular potential well at the Si/SiO2 interface narrower and deeper. Just like a particle-in-a-box, a narrower confinement well forces all the quantized energy levels to shift to higher energies and also pushes them further apart.

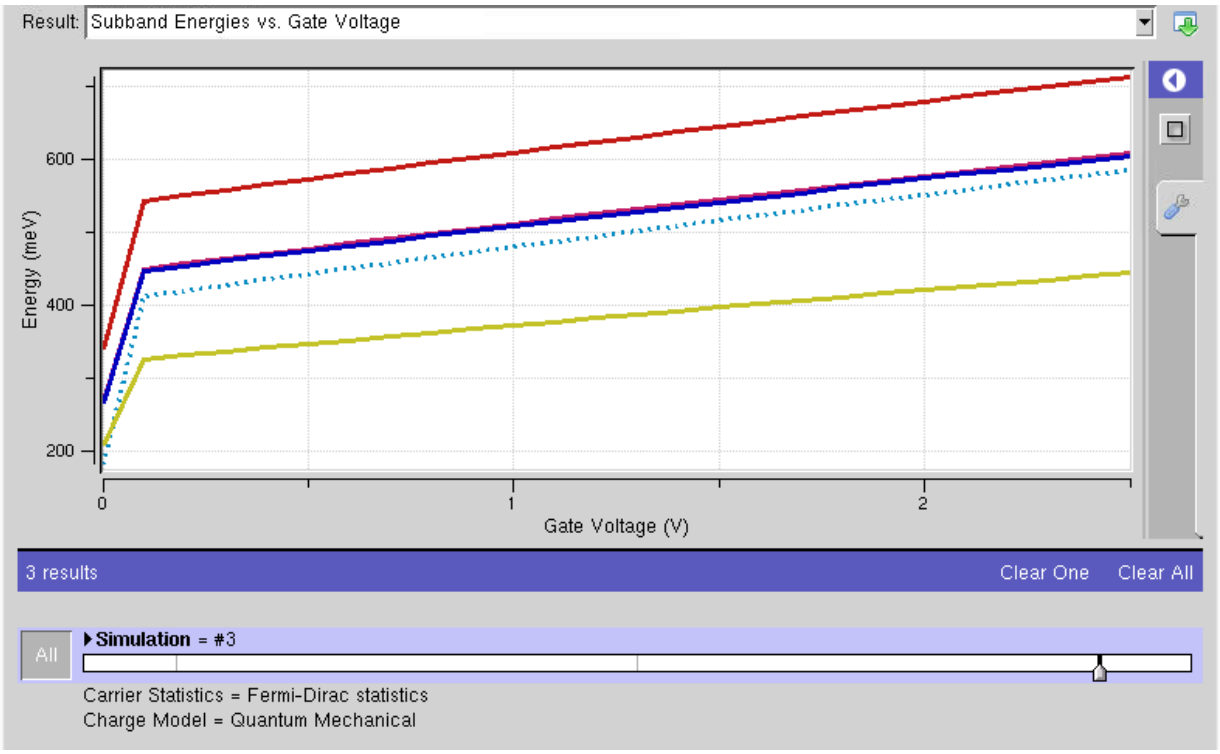
Simulation 1:



Simulation 2:



Simulation 3:

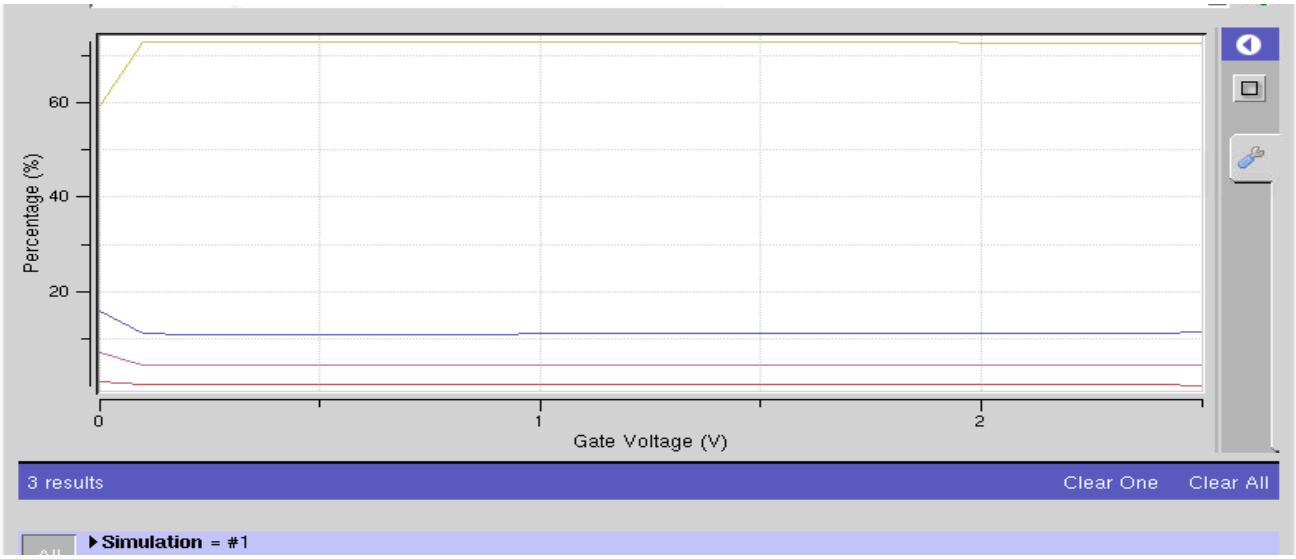


F. For the quantum-mechanical charge description model, plot the subband population as a function of the gate voltage. What subband is the most occupied one? How does the variation of the occupancy of the most occupied subband vary with the gate voltage? Why?

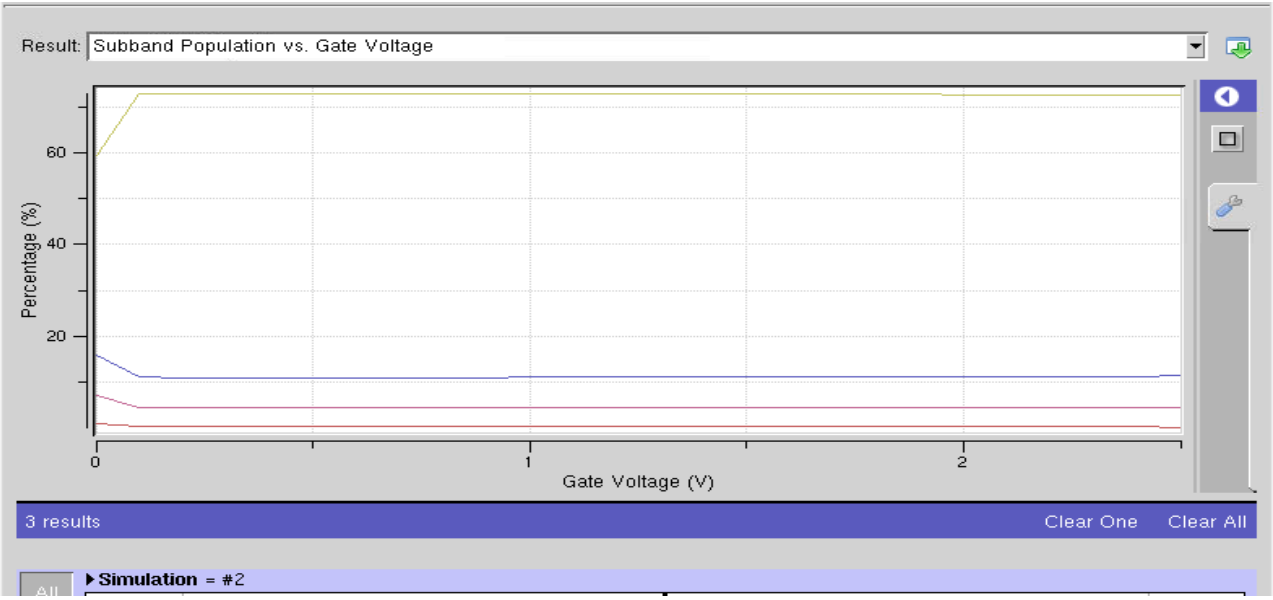
Answer: Based on my analysis of this data, the most occupied subband is P13, which corresponds to the ground state of the unprimed (Delta_2) ladder. At 0V, it already holds 59.4% of the electrons, and this occupancy quickly rises to a peak of 72.95% at 0.7V.As the gate voltage increases from 0.7V to 2.5V, the occupancy of this ground state slowly decreases from 72.95% down to 72.36%. At the same time, the populations of the higher-energy subbands, like P11 (the next unprimed state) and P23 (the ground state of the primed ladder), show a slight increase. This happens because as the gate voltage increases, all the

subband energy levels move further away from the Fermi level, but the Fermi level itself also moves to a higher energy (from 412.78 meV at 0.1V to 586.37 meV at 2.5V). This rise in the Fermi level makes it statistically more likely (via Fermi-Dirac statistics) for electrons to populate the higher subbands, causing a slight redistribution of electrons out of the ground state and into these higher states.

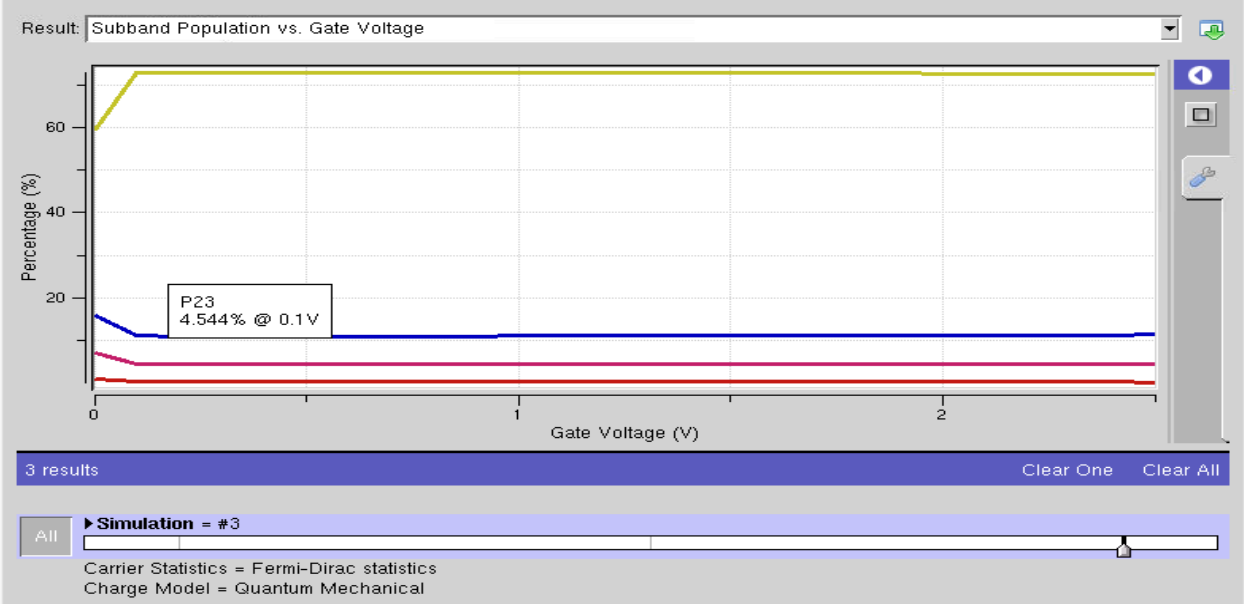
Simulation1:



Simulation 2:



Simulation 3:



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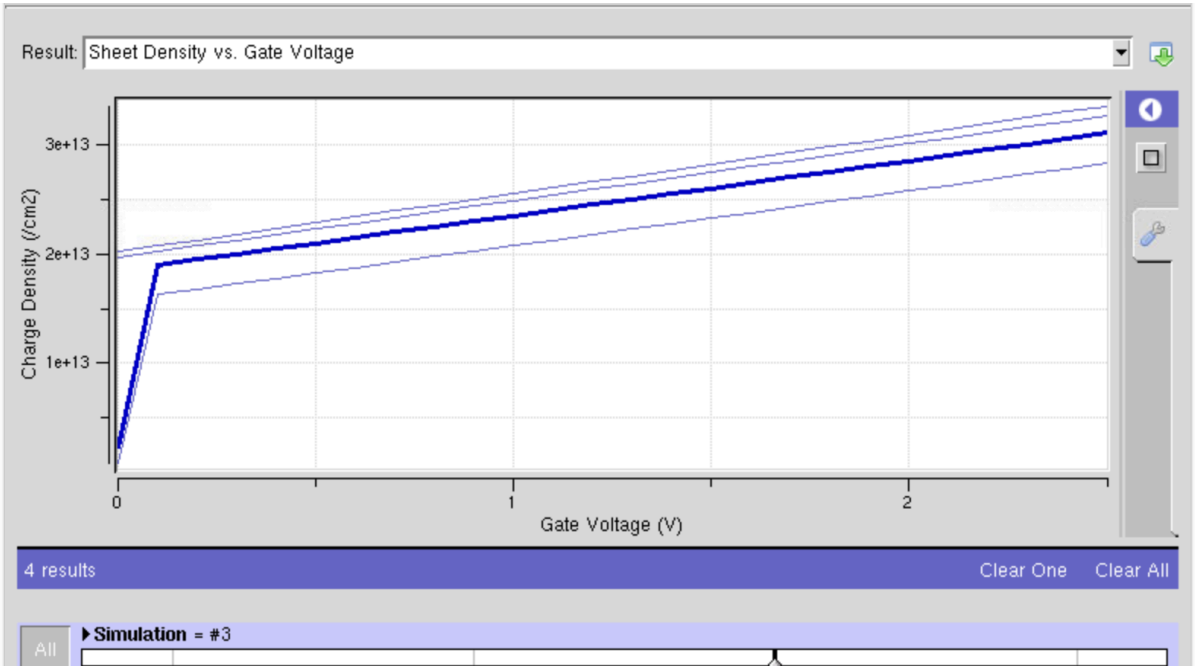
2. *(2 points)* Run Simulation setups 3 and 4 and answer the following questions:

A. Plot the electron sheet density as a function of gate voltage for the two simulation setups. What can you say about the sheet density variation (and the threshold voltage) as a function of the substrate doping?

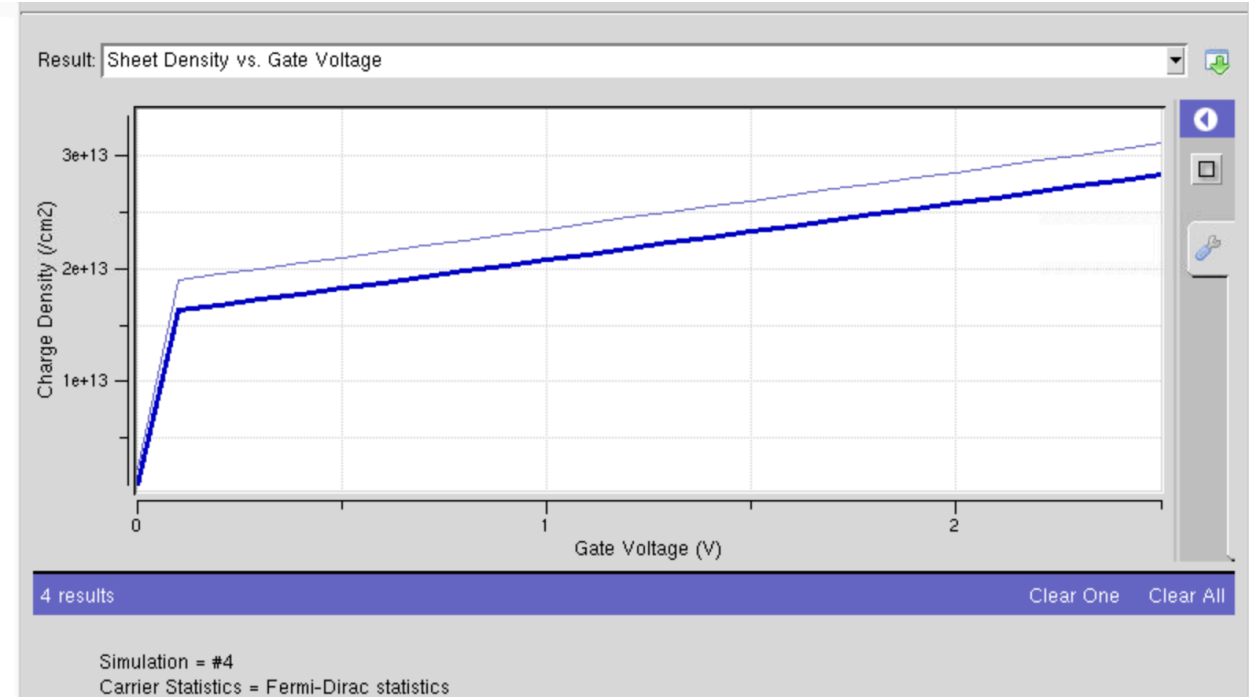
Answer: I plotted the electron sheet density for the two quantum-mechanical simulations: Simulation 3 (10^{17} cm^{-3} doping) and Simulation 4 (10^{18} cm^{-3} doping). The data clearly shows that the simulation with the higher substrate doping (Simulation 4) has a significantly lower sheet density for any given gate voltage. For example, at 2.5V, the sheet density for Simulation 4 is 2.83 x 10^{13} /cm^2, which is lower than the 3.11 x 10^{13} /cm^2 from Simulation 3. This indicates that the threshold voltage increases with higher substrate doping. This is because a higher doping concentration creates a stronger built-in electric field that the gate

voltage must overcome. Furthermore, the higher doping makes the triangular potential well at the surface narrower, increases the quantum confinement, raises the energy of the ground state subband, and therefore further increases the threshold voltage .

Simulation 3:



Simulation 4:

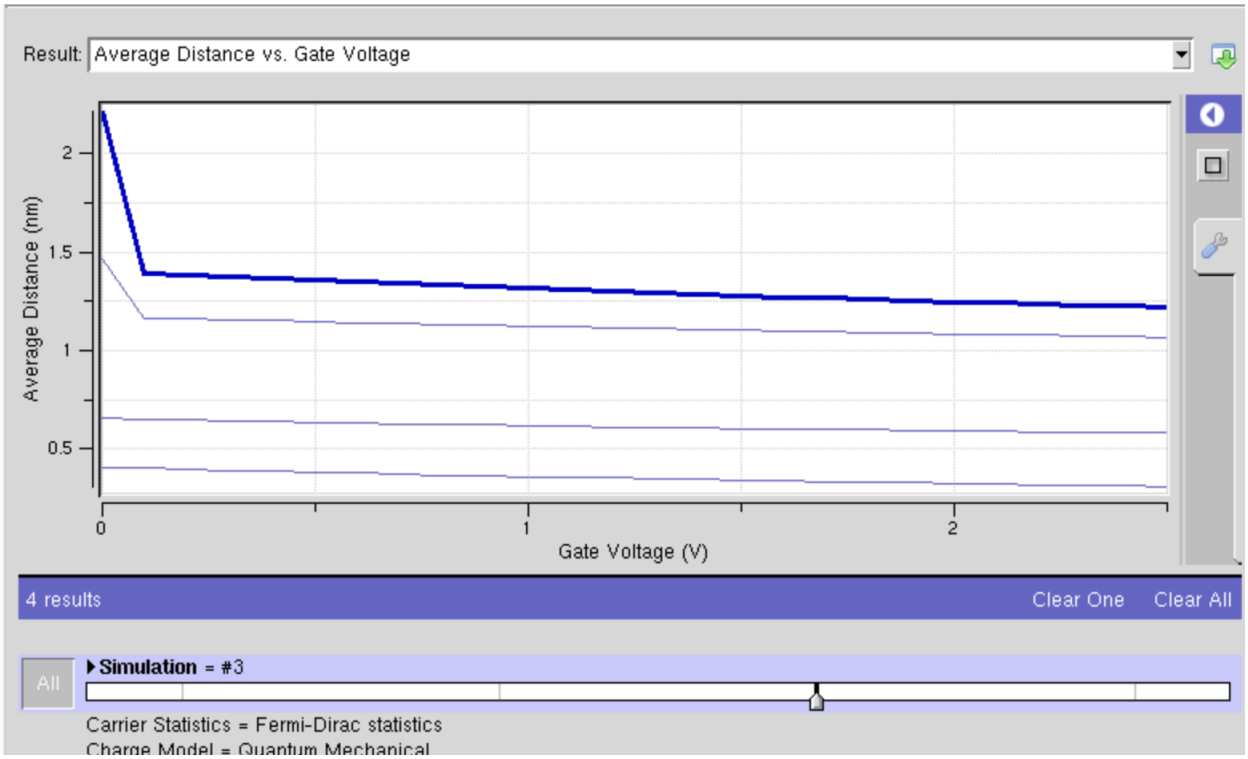


B. Plot the average distance of the electrons from the interface for the two simulation setups. Given the gate voltage of 2V, under what conditions we have minimum average distance of the electrons from the interface. Why?

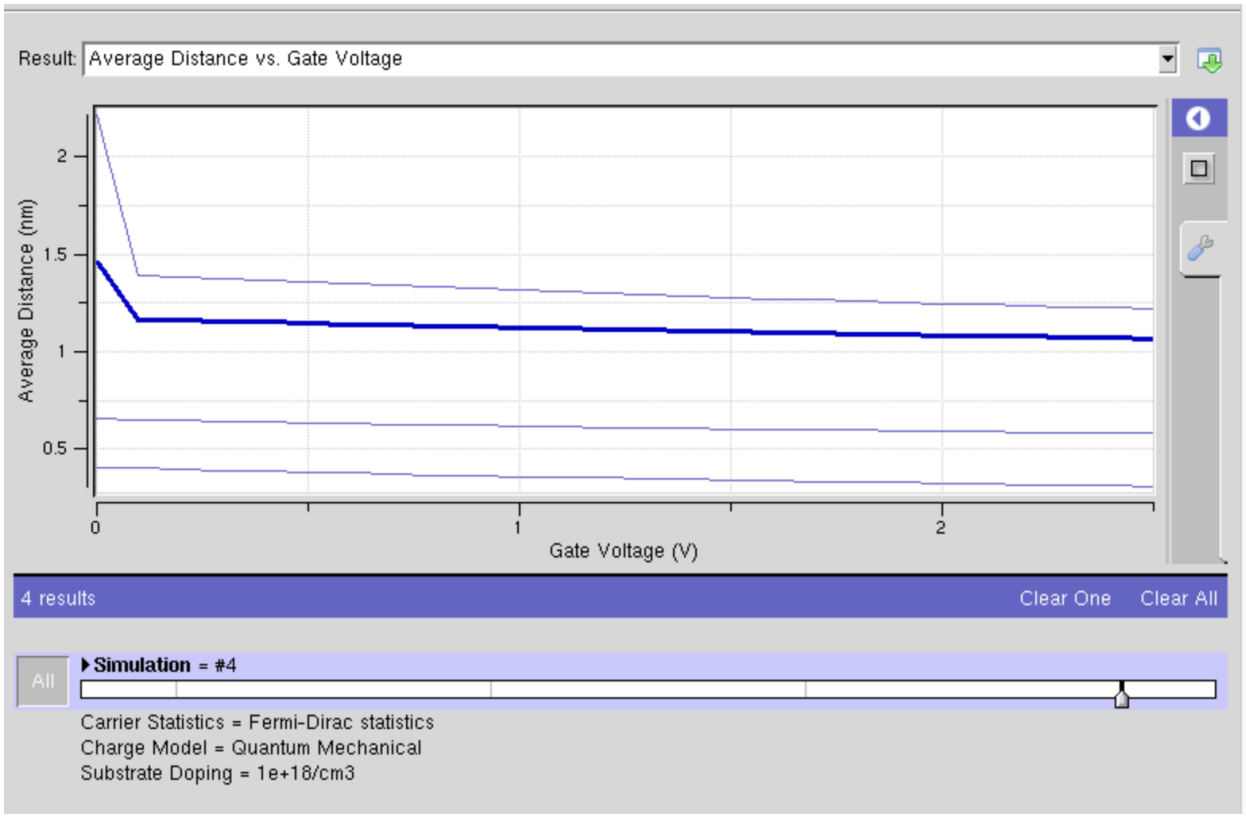
Answer: I plotted the average distance of the electrons from the interface for the two quantum-mechanical simulations: Simulation 3 (10^{17} cm^{-3} doping) and Simulation 4 (10^{18}

cm^{-3} doping). The simulation data shows that at a gate voltage of 2V, the minimum average distance of 1.084 nm occurs in Simulation 4, which is the case with the higher substrate doping (10^{18} cm^{-3}). This is less than the 1.247 nm distance from Simulation 3 (10^{17} cm^{-3}). This happens because the higher acceptor doping (N_A) in Simulation 4 creates a stronger initial electric field at the interface. This, in turn, creates a narrower, deeper triangular potential well, which confines the electrons more tightly against the interface and reduces their average distance.

Simulation 3:



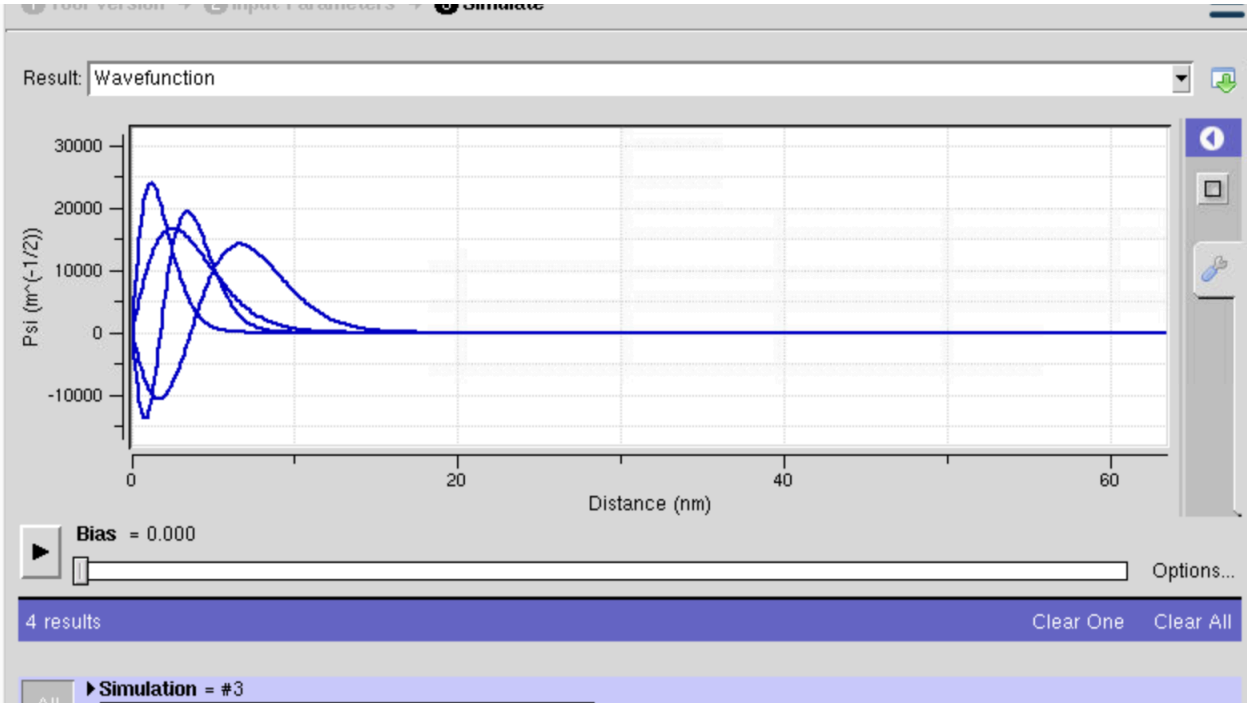
Simulation 4:



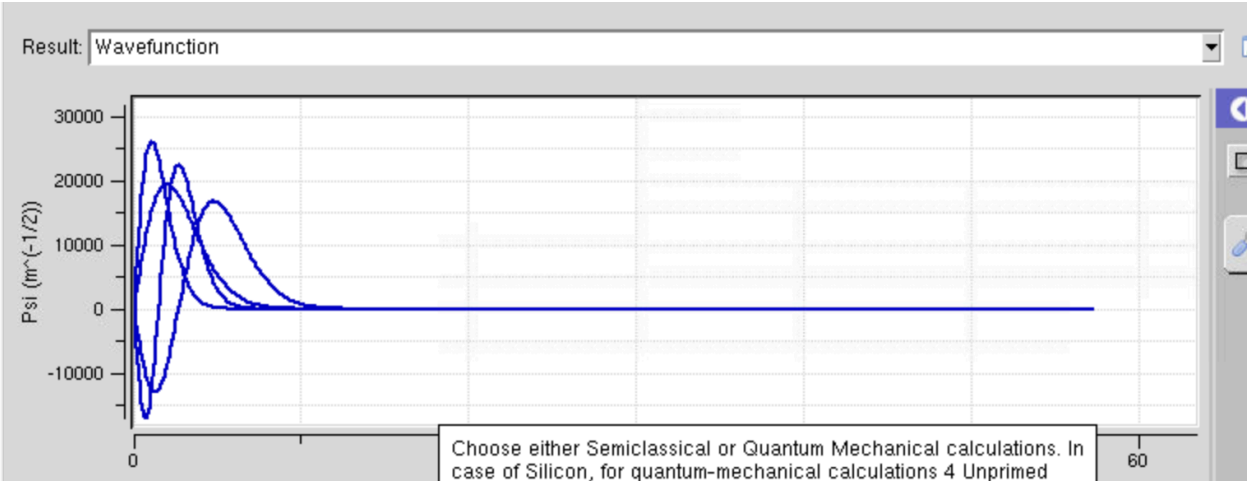
C. For the quantum-mechanical charge description model, plot the unprimed and primed wavefunctions for V_G=2V. What can you say about the spatial extent of these wavefunctions for the electrons sitting in the unprimed or primed ladder of subbands as a function of the substrate doping?

Answer: My analysis of the data shows that the wavefunctions in the higher-doped simulation (Simulation 4, 10^{18} cm^{-3}) are more spatially confined and are pulled closer to the Si/SiO_2 interface.This is clear from comparing the peak positions of the ground state wavefunctions. In the low-doping case (Simulation 3, 10^{17} cm^{-3}), the ground state of the unprimed ladder (psi11) peaks around 2.4-2.7 nm away from the interface. In the high-doping case (Simulation 4, 10^{18} cm^{-3}), this peak is pulled in to about 1.5-1.65 nm. This same trend is visible in the primed ladder; the ground state (psi21) peak moves from around 6.3-6.75 nm in the low-doping case to 5.4 nm in the high-doping case.This is a direct result of the stronger electric field from the higher acceptor doping (N_A). This stronger field creates a narrower and steeper triangular potential well at the surface, which squeezes the electron wavefunctions, confining them more tightly and pulling their probability peaks closer to the interface 2222.

Simulation 3:



Simulation 4:

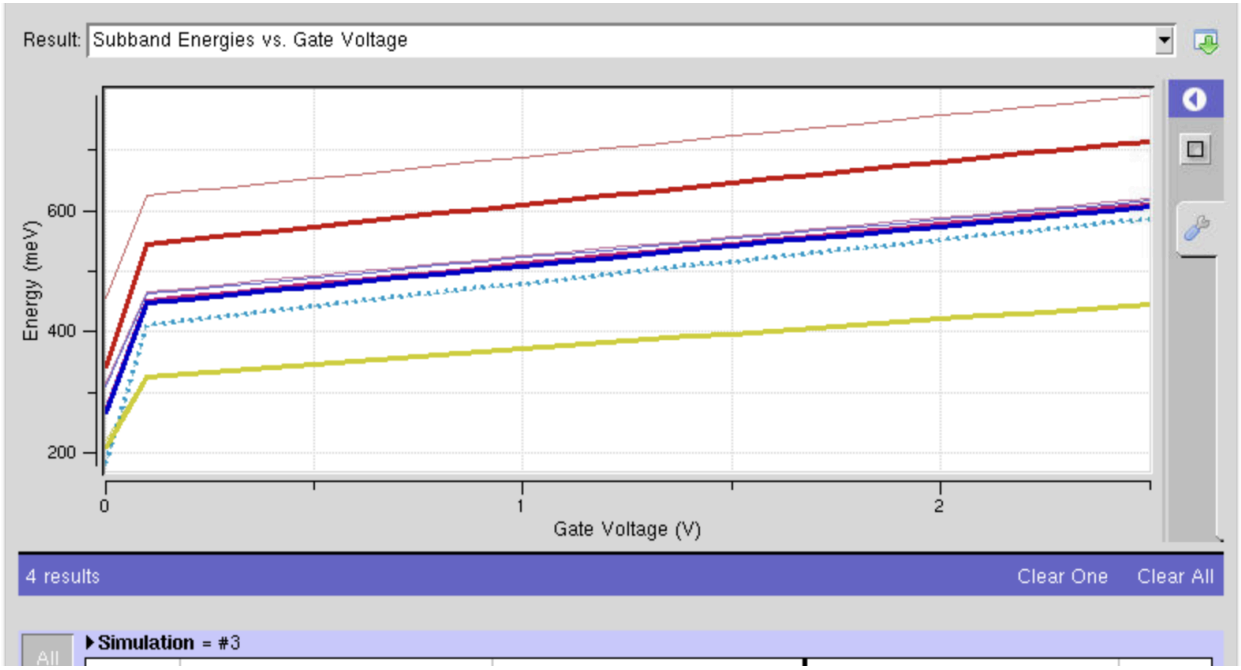


D. For the quantum-mechanical charge description model, plot the subband energies vs gate voltage? What happens with the subband energies and the subband separation when increasing the substrate doping? Why?

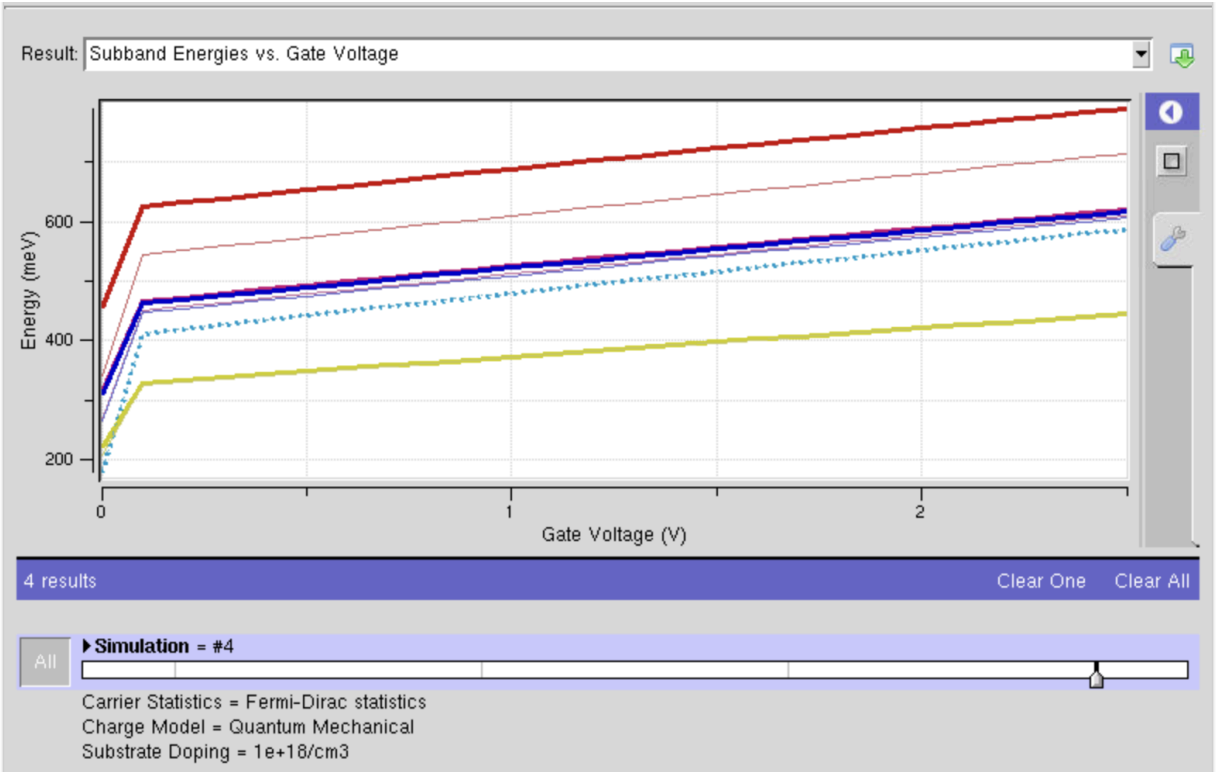
Answer: I plotted the subband energies for both quantum-mechanical simulations: Simulation 3 (10^{17} cm^{-3} doping) and Simulation 4 (10^{18} cm^{-3} doping). My analysis of the data shows that increasing the substrate doping shifts all subband energies to higher values and also increases the separation between the subbands.For example, at a gate voltage of 2V, the ground state energy (E11) is 573.87 meV for the low-doping case (Sim 3) but increases to 585.49 meV for the high-doping case (Sim 4). Similarly, the separation between the first two unprimed subbands (E21 - E11) at 2V is 106.29 meV (680.16 - 573.87) for Sim 3, but this separation increases to 170.89 meV (756.38 - 585.49) for Sim 4.This is because the higher acceptor doping (N_A) in Simulation 4 creates a stronger electric field at the surface. This

stronger field results in a narrower and steeper triangular potential well. Just as with the effect of increasing gate voltage, this stronger spatial confinement forces all the quantized energy levels to shift to higher energies and also pushes them further apart from each other 1.

Simulation 3:



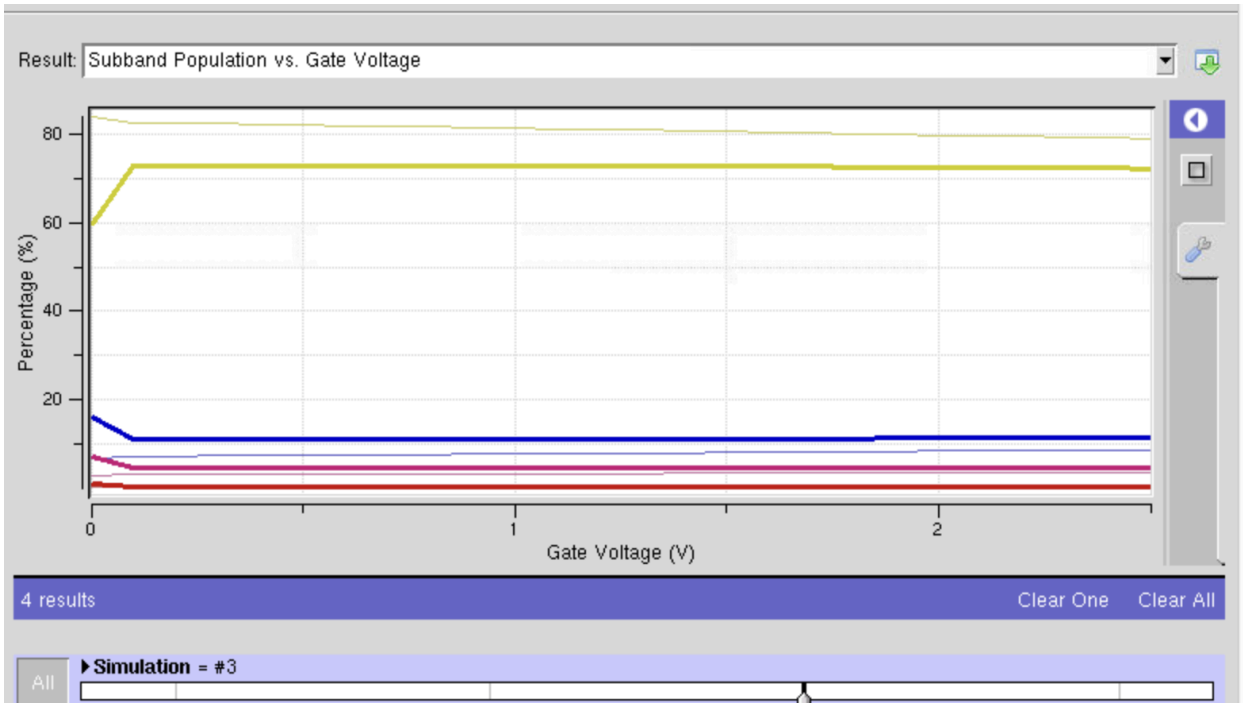
Simulation 4:



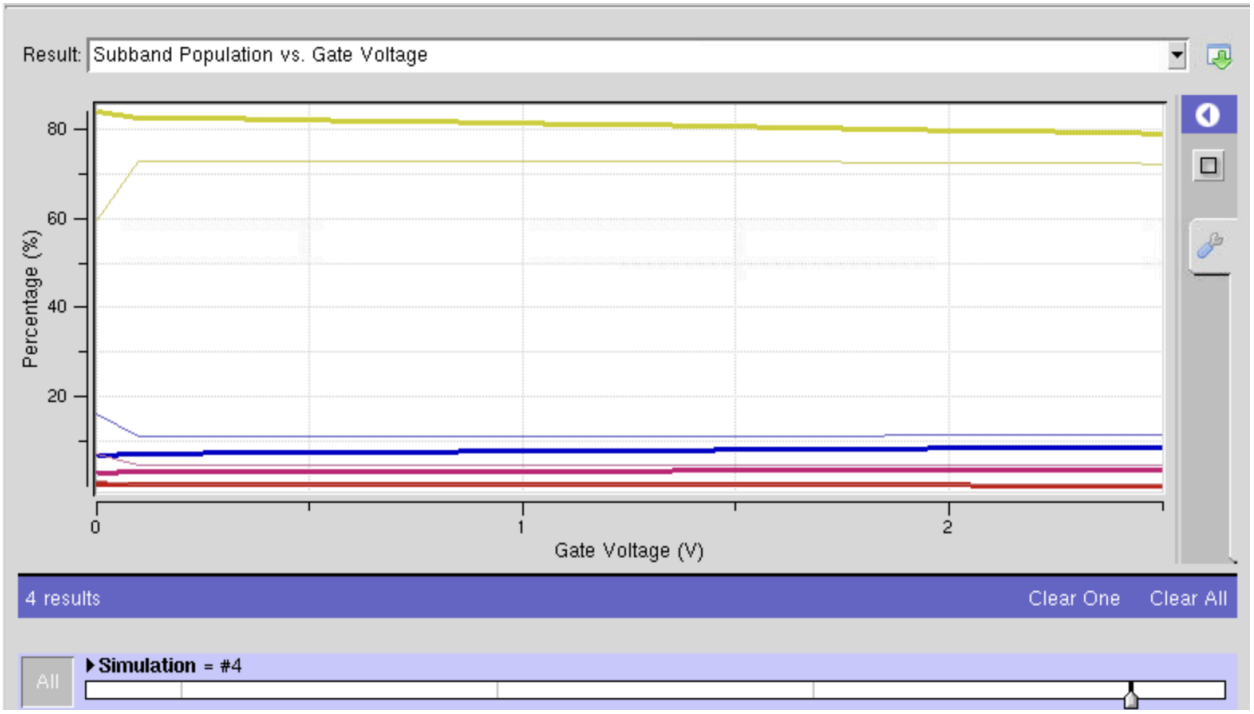
E. For the quantum-mechanical charge description model, plot the subband population as a function of the gate voltage. What subband is the most occupied one? How does the variation of the occupancy of the most occupied subband vary with the substrate doping? Why?

Answer: I plotted the subband population as a function of gate voltage for both Simulation 3 (10^{17} cm^{-3} doping) and Simulation 4 (10^{18} cm^{-3} doping). In both cases, the data clearly shows that the subband labeled P13 is the most occupied one. This subband corresponds to the ground state of the unprimed (Delta_2) ladder, which is the lowest energy state available.When comparing the two simulations, the occupancy of this ground state (P13) is significantly higher in the high-doping simulation (Sim 4) than in the low-doping one (Sim 3). At a gate voltage of 2.5V, the P13 occupancy is 79.15% for the high-doping case, but only 72.36% for the low-doping case. The reason for this, which I also found from my subband energy data, is that the higher substrate doping in Simulation 4 creates a stronger electric field and a narrower, steeper potential well. This increased quantum confinement increases the energy separation between the subbands 1. Because the energy gap between the ground state (P13) and the next-lowest states (like P11) is larger in the high-doping case, it is statistically less probable for electrons to have enough thermal energy to populate those higher states. This results in a larger percentage of the total electron population being confined to the ground state, pushing the device further into the electronic quantum limit 2.

Simulation 3:



Simulation 4:



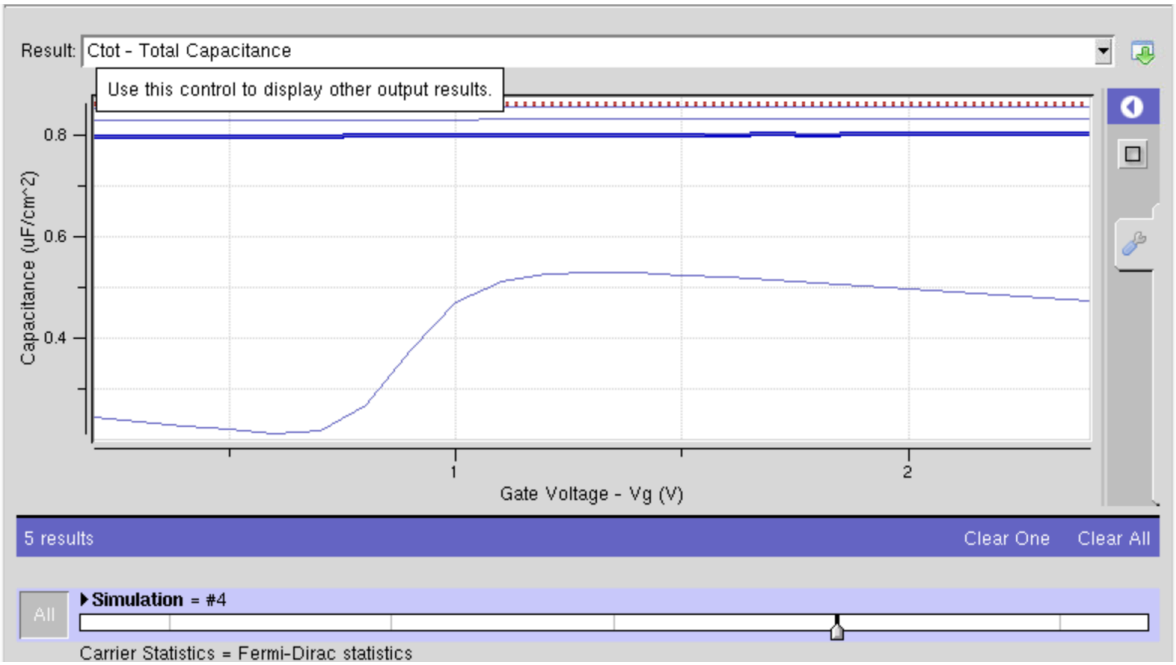
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3. *(1.2 points)* Run Simulation setups 4 and 5 and answer the following questions:

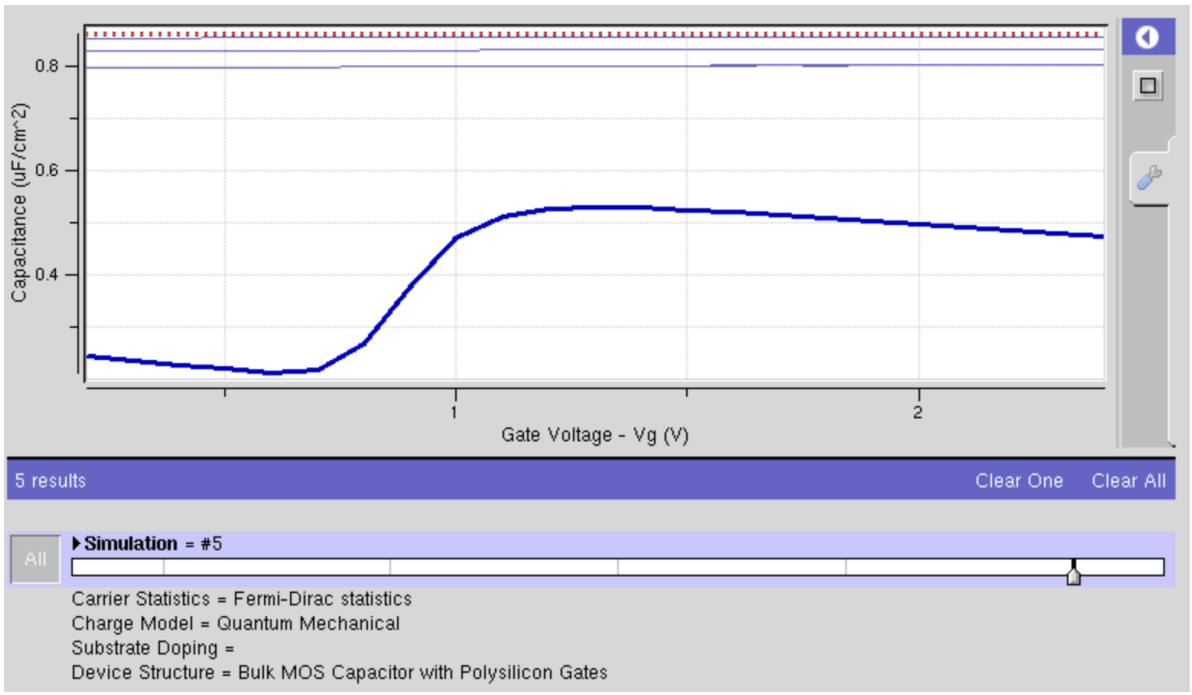
A. Plot the total gate capacitance as a function of the gate voltage. For the gate voltage of 2V what is the degradation of the total gate capacitance when using quantum mechanical model and metal and polysilicon gates? What is the gate voltage dependence of the total gate capacitance under inversion bias conditions?

Answer: I plotted the total gate capacitance (Ctotal) for Simulation 4 (quantum model with metal gate) and Simulation 5 (quantum model with polysilicon gate). To find the degradation at 2V, I compared the Ctotal of each to the ideal oxide capacitance (Cox), which my results show as 0.864 µF/cm² for both simulations. For the metal gate (Simulation 4), the Ctotal at 2V was 0.803 µF/cm², which corresponds to a degradation of 7.06%. For the polysilicon gate (Simulation 5), the Ctotal at 2V was 0.497 µF/cm², showing a much more severe degradation of 42.48%. Regarding the gate voltage dependence under inversion bias, the two models behave very differently. The Ctotal for the metal gate (Simulation 4) is relatively flat in the inversion region, staying almost constant around 0.800-0.804 µF/cm². In sharp contrast, the Ctotal for the polysilicon gate (Simulation 5) shows a pronounced gate-voltage dependence: after peaking at 0.53 µF/cm² at 1.3V, the capacitance decreases steadily as the gate voltage increases further, dropping to 0.473 µF/cm² by 2.4V. Strong gate voltage causes a depletion region to form within the poly-gate itself, which acts as an additional, voltage-dependent capacitor in series, significantly degrading the total capacitance.

Simulation4:



Simulation 5:

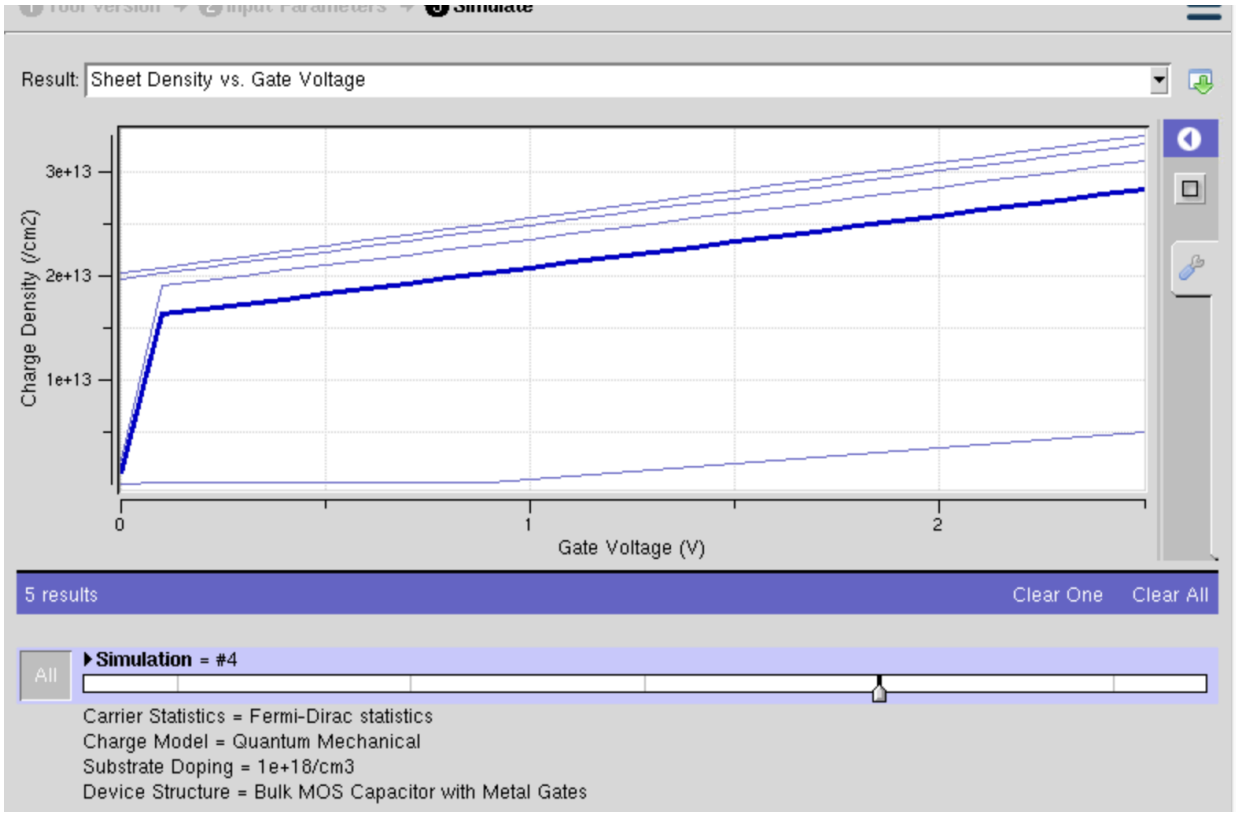


B. Plot the electron sheet density as a function of gate voltage for the two simulation setups. What can you say about the sheet density variation (and the threshold voltage) for the case of metal and polysilicon gates?

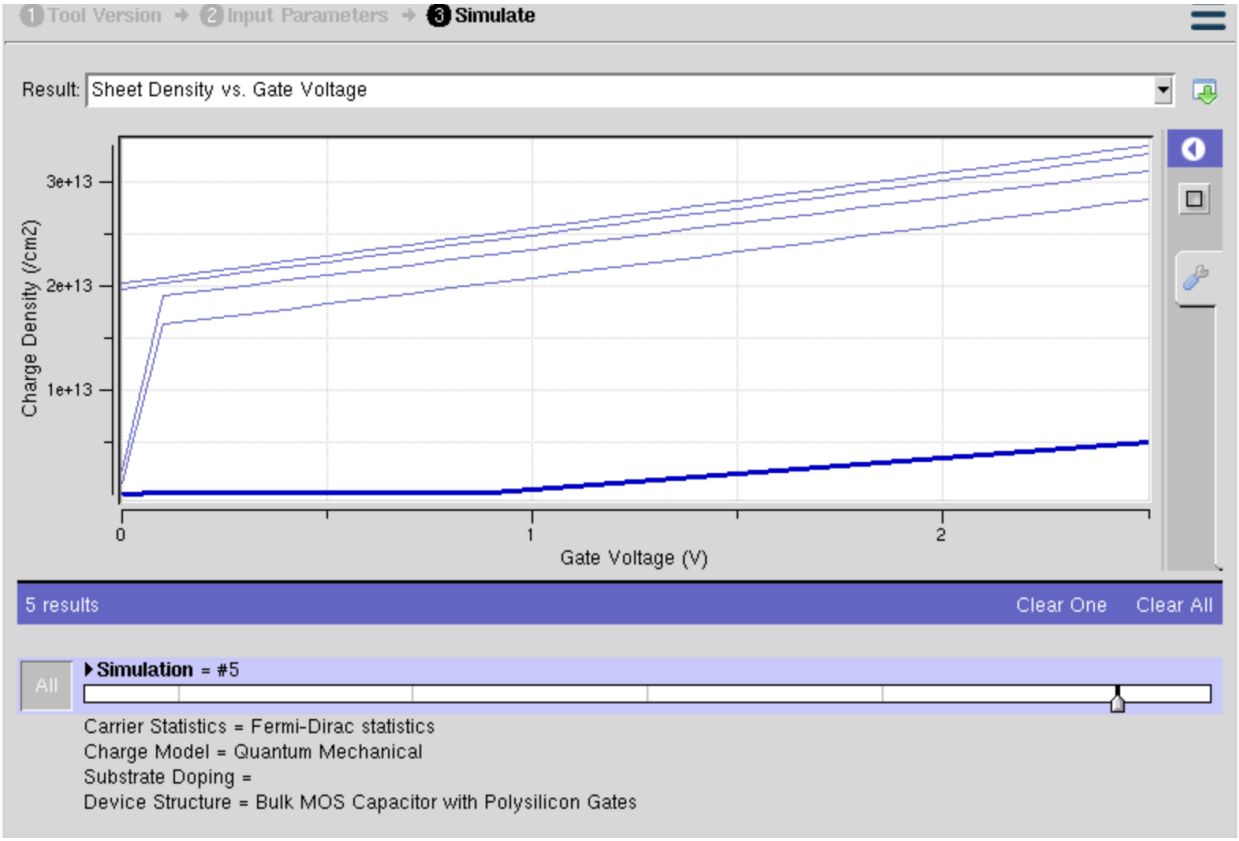
Answer: I plotted the electron sheet density for Simulation 4 (metal gate) and Simulation 5 (polysilicon gate). My simulation data shows a dramatic difference between the two. The metal gate case (Simulation 4) shows a high sheet density that increases linearly, starting at 1.68 x 10^{13} /cm² at 0.2V and rising to 2.83 x 10^{13} /cm² at 2.5V. The polysilicon gate

(Simulation 5), however, has a much lower sheet density, starting near zero (only 8033 /cm² at 0.2V) and only beginning to increase significantly after 0.8V. This indicates the threshold voltage for the polysilicon gate is much higher (around 0.8V-0.9V) than for the metal gate (which is below 0.2V). This is a direct consequence of the poly-gate depletion effect. As my lecture materials explain, a depletion region forms inside the polysilicon gate itself, which causes a voltage drop. This means a larger external gate voltage must be applied just to overcome this internal voltage drop before an inversion layer can even begin to form in the channel .

Sim 4:



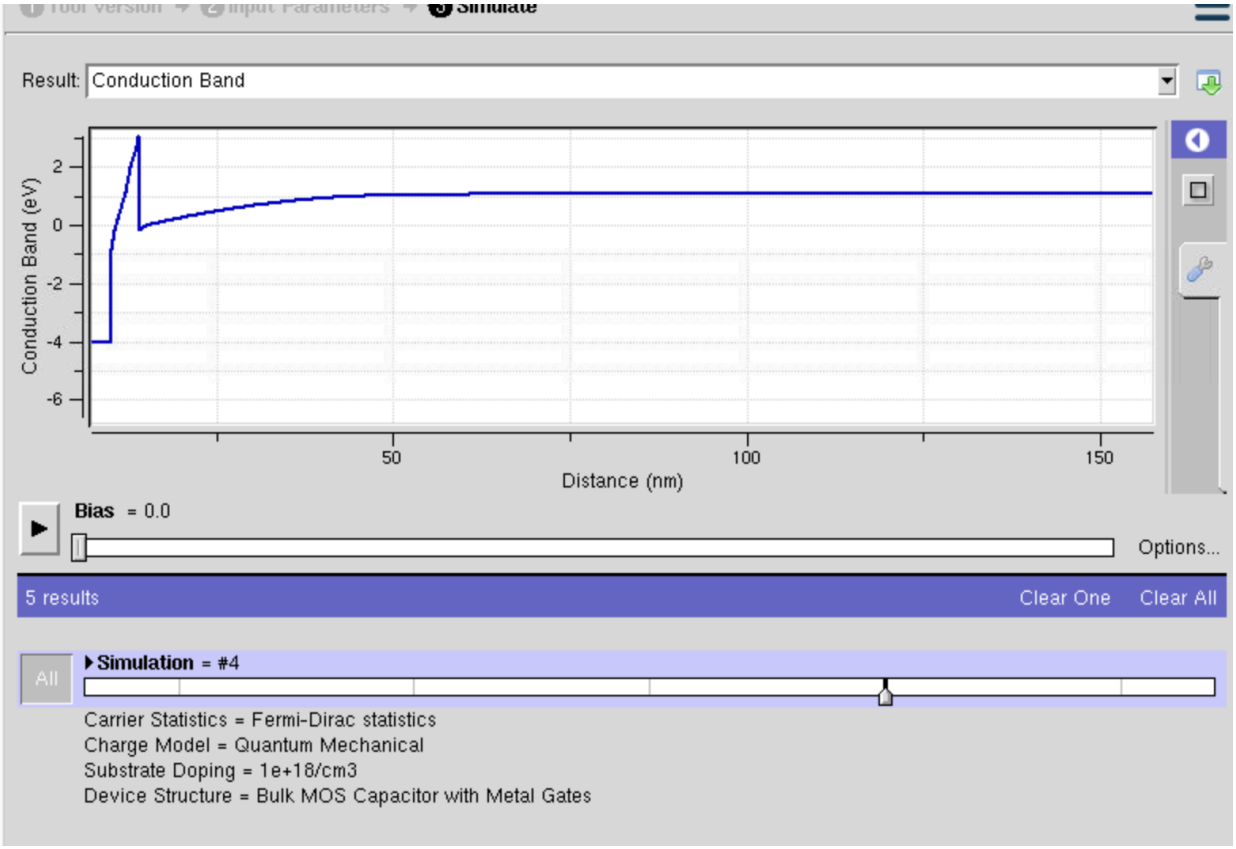
Sim 5:



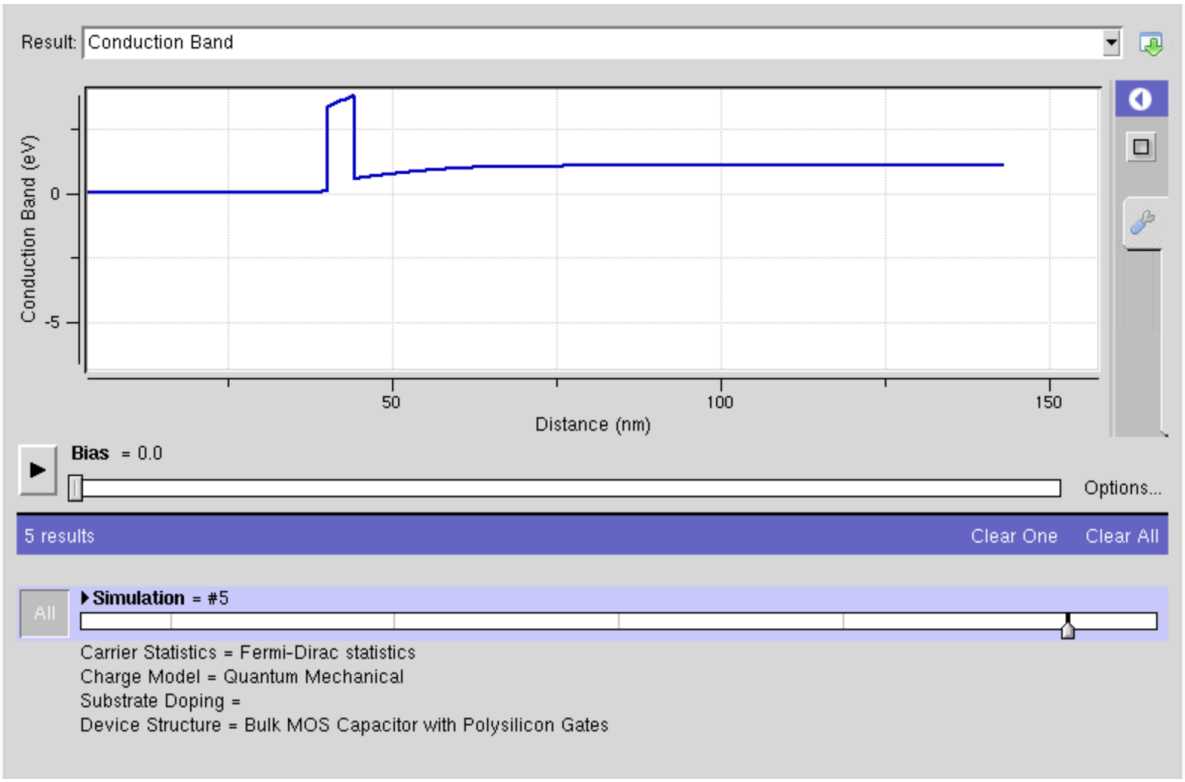
C. Plot the conduction band profile for the case of quantum mechanical models with metal and polysilicon gates. How significant is the depletion of the polysilicon gates for V_G=2.5 V.

Answer:The depletion of the polysilicon gate is clearly visible and significant. In the conduction band plot for the metal gate, the energy level in the gate region (distances less than 10 nm) is perfectly flat, as expected for a metal with no internal voltage drop. However, in the plot for the polysilicon gate, the conduction band in the gate region (distances less than 40 nm) is not flat; it clearly bends upward as it approaches the oxide. Based on my simulation data for VG =2.5 V, the conduction band energy at the far end of the poly-gate (at 3.5 nm) is 0.023 eV, while the energy at the poly-oxide interface (at 40.02 nm) has risen to 0.096 eV. This band bending reveals an internal voltage drop of 0.073 V across the polysilicon gate itself. This is the poly-depletion effect, and it is significant because this voltage is "lost" and does not contribute to bending the bands in the silicon channel, which, as I saw in my previous results, increases the threshold voltage and severely degrades the total gate capacitance .

Sim 4:

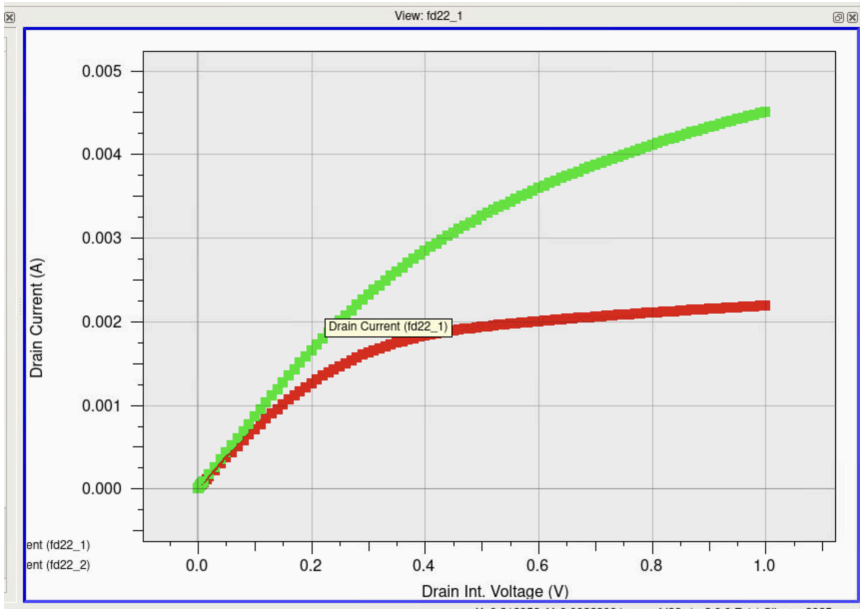

Result: Conduction Band
Conduction Band (eV)
2
0
-2
-4
-6
50
100
150
Distance (nm)
Bias = 0.0
Options...
5 results
Clear One
Clear All
All
Simulation = #4
Carrier Statistics = Fermi-Dirac statistics
Charge Model = Quantum Mechanical
Substrate Doping = 1e+18/cm3
Device Structure = Bulk MOS Capacitor with Metal Gates

Sim 5:


Result: Conduction Band
Conduction Band (eV)
0
-5
50
100
150
Distance (nm)
Bias = 0.0
Options...
5 results
Clear One
Clear All
All
Simulation = #5
Carrier Statistics = Fermi-Dirac statistics
Charge Model = Quantum Mechanical
Substrate Doping =
Device Structure = Bulk MOS Capacitor with Polysilicon Gates

Part 2: Single and Dual Gate Quantum SOI Devices Modeling With Silvaco

Use the same input script as in SOI Lab 3 for modeling a 28 nm channel length SOI device structure. Create a dual gate structure as depicted on the figure. The two gate oxide thicknesses are the same. The top and bottom gates have to be connected with the parameter **COMMON** on the CONTACT statement as they are biased at the same level. *Assume isothermal energy balance model for the carriers' transport*. Perform the following simulations:



(a) (*3 points*) Simulate the **single-gate** structure (calculate output I_D-V_D characteristic for V_G=V_{th}+0.7 V). Then simulate the **dual-gate** structure. Plot the I_D-V_D characteristics for the single and the dual gate structure on the same plot. Register the ON-currents at V_D=0.9 V. What is the ratio of the two ON currents?

I simulated the output characteristics, which is drain current versus drain voltage, for both the 28 nm single-gate and dual-gate SOI structures using the semiclassical model. I biased the gates at a gate voltage equal to the threshold voltage plus 0.7 V. After plotting both curves on the same graph, I registered the ON-currents at a drain voltage of 0.9 V. My results were 0.01411 A for the single-gate device and 0.02470 A for the dual-gate device. The dual-gate device clearly shows a much stronger drive current. The ratio of the two ON-currents, which I calculated as the dual-gate current divided by the single-gate current, is 1.750.

b) (*2.4 points*) Consider the single-gate and the dual-gate structures from part (a). Assume V_D=0.05 V. Ramp the gate voltage to calculate the transfer characteristics. Use in one case semi-classical charge description and in second case *Bohm Quantum Potential* (**BQP.N**) for the electrons. Calculate the threshold voltages. What are the shifts of the threshold voltages due to quantum-mechanical size-quantization effects? Are they more pronounced in the single-gate or the dual-gate structure?

Next, I simulated the transfer characteristics, or drain current versus gate voltage, for both the single-gate and dual-gate structures at a drain voltage of 0.05 V. I ran these simulations twice: once with the standard semiclassical charge description and a second time using the Bohm Quantum Potential model for the electrons. Using a constant-current method, I extracted the following threshold voltages: 0.0458 V for the semiclassical single-gate, 0.1753 V for the quantum single-gate, -0.0130 V for the semiclassical dual-gate, and 0.0505 V for the quantum dual-gate. From these values, I calculated the shifts in the threshold voltage due to the quantum-mechanical size-quantization effects. The shift for the single-gate device was an increase of 0.1295 V, while the

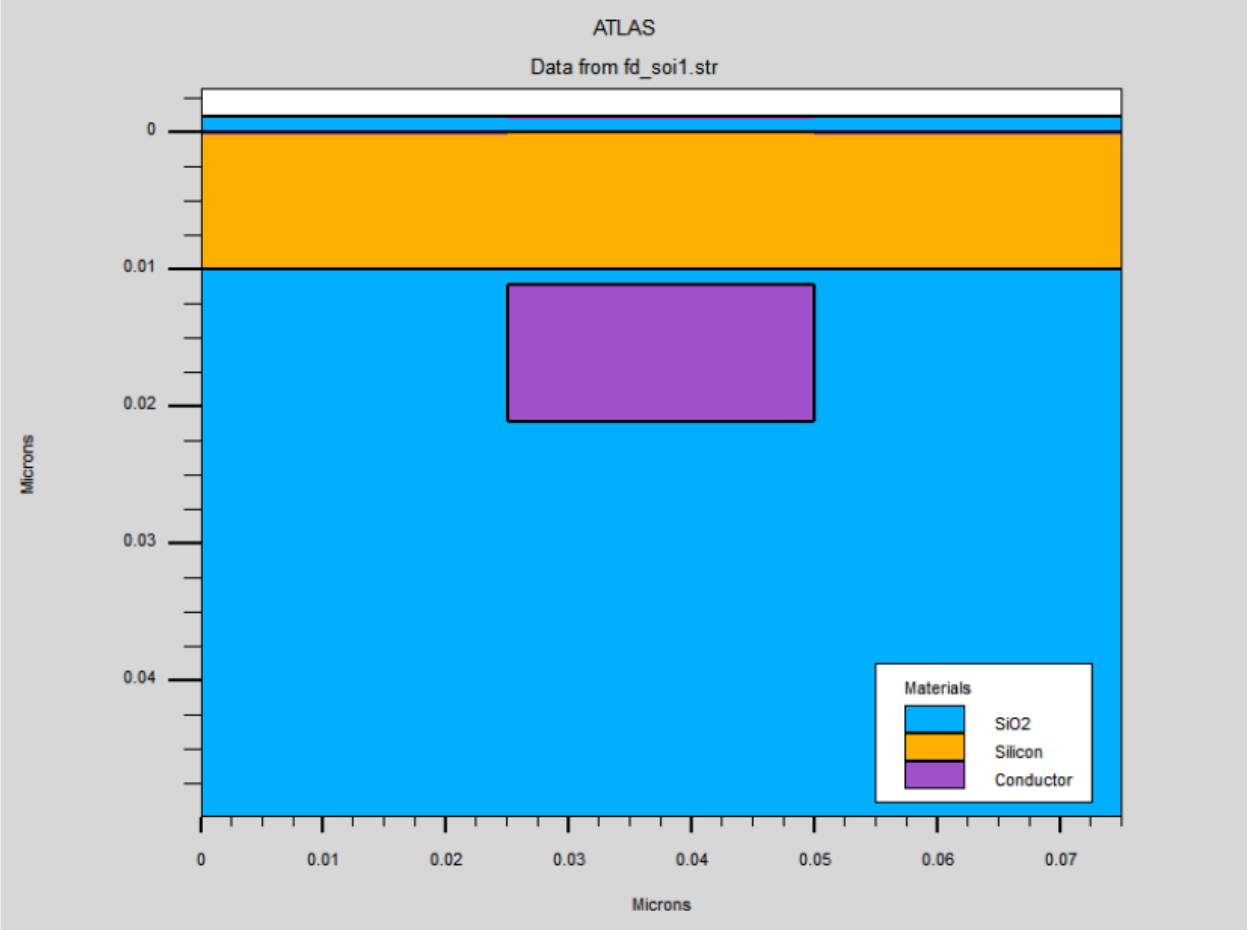
shift for the dual-gate device was an increase of 0.0635 V. These results clearly show that the quantum-mechanical effects are more pronounced in the single-gate structure, which had a threshold voltage shift about twice as large as the dual-gate structure.

c) (*1 point*) Solve for the output characteristics as in part (a), for single and dual-gate SOI device structures, but now include the *quantum correction effects using the Bohm Quantum Potential* (**BQP.N**) for the electrons. Comment on the ratio of the ON currents (ratio of the on-current for single-gate and dual-gate SOI device structure with and without Bohm Quantum Potential).
Finally, I solved for the output characteristics again, but this time I included the quantum correction effects using the Bohm Quantum Potential for both the single-gate and dual-gate SOI device structures. I registered the new ON-currents at a drain voltage of 0.9 V and found them to be 0.01588 A for the single-gate device and 0.02117 A for the dual-gate device. This gives a new ON-current ratio (dual-gate divided by single-gate) of 1.333. When I comment on the ratio, I see that the performance advantage of the dual-gate structure decreased when quantum effects were included. The ratio dropped from 1.750 in the semiclassical case down to 1.333 in the quantum-corrected case. This implies that quantum effects impact the device performance in a way that reduces the benefit of the dual-gate design.

For completeness, here are the parameters of the 28 nm channel length SOI device structure:

- Oxide thickness = 1.2 nm .
- Silicon film thickness = 7 nm .
- For the single-gate SOI device structure, the BOX thickness is 25 nm .
- The doping of the Si-film is 5×10^{15} cm^{-3}.
- Gate is aluminum.

The bottom gate oxide thickness of the dual gate structure is also 1.2 nm. Be careful with meshing when specifying the bottom gate.



Schematic of the dual-gate structure.